

BOOK 9

PARTICLE PHYSICS

11 Particle physics

11.1 Atoms, nuclei and radiation

Candidates should be able to:

- 1 infer from the results of the α -particle scattering experiment the existence and small size of the nucleus
- 2 describe a simple model for the nuclear atom to include protons, neutrons and orbital electrons
- 3 distinguish between nucleon number and proton number
- 4 understand that isotopes are forms of the same element with different numbers of neutrons in their nuclei
- 5 understand and use the notation A_ZX for the representation of nuclides
- 6 understand that nucleon number and charge are conserved in nuclear processes
- 7 describe the composition, mass and charge of α -, β - and γ -radiations (both β^- (electrons) and β^+ (positrons) are included)
- 8 understand that an antiparticle has the same mass but opposite charge to the corresponding particle, and that a positron is the antiparticle of an electron
- 9 state that (electron) antineutrinos are produced during β^- decay and (electron) neutrinos are produced during β^+ decay
- 10 understand that α -particles have discrete energies but that β -particles have a continuous range of energies because (anti)neutrinos are emitted in β -decay
- 11 represent α - and β -decay by a radioactive decay equation of the form ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^4_2\alpha$
- 12 use the unified atomic mass unit (u) as a unit of mass

11.2 Fundamental particles

Candidates should be able to:

- 1 understand that a quark is a fundamental particle and that there are six flavours (types) of quark: up, down, strange, charm, top and bottom
- 2 recall and use the charge of each flavour of quark and understand that its respective antiquark has the opposite charge (no knowledge of any other properties of quarks is required)
- 3 recall that protons and neutrons are not fundamental particles and describe protons and neutrons in terms of their quark composition
- 4 understand that a hadron may be either a baryon (consisting of three quarks) or a meson (consisting of one quark and one antiquark)
- 5 describe the changes to quark composition that take place during β^- and β^+ decay
- 6 recall that electrons and neutrinos are fundamental particles called leptons

1 9702/1/M/J/02/Q38

Which set of radioactive emissions corresponds to the descriptions given in the table headings?

	high-speed electrons	high-speed helium nuclei	high-frequency photons
A	α	β	γ
B	α	γ	β
C	β	α	γ
D	β	γ	α

2 9702/1/M/J/02/Q39

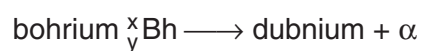
The nucleus of one of the isotopes of nickel is represented by ${}_{28}\text{Ni}$.

Which line in the table correctly describes a neutral atom of this isotope?

	number of protons	number of neutrons	number of orbital electrons
A	28	32	28
B	28	60	28
C	60	28	28
D	60	32	32

3 9702/1/M/J/02/Q40

A nucleus of bohrium ${}^x_y\text{Bh}$ decays to mendelevium ${}^{255}_{101}\text{Md}$ by a sequence of three α -particle emissions.



How many neutrons are there in a nucleus of ${}^X_Y\text{Bh}$?

- A** 267 **C** 160 **B** 261 **D** 154

4 9702/1/O/N/02/Q38

The numbers of protons, neutrons and nucleons in three nuclei are shown.

nucleus	number of protons	number of neutrons	number of nucleons
X	15	16	3
Y	15	17	3
Z	16	16	3

Which nuclei are isotopes of the same element?

- A** X and Y **B** X and Z **C** Y and Z **D** none of them

5 9702/1/O/N/02/Q39

In an experiment to investigate the nature of the atom, a very thin gold film was bombarded with α -particles.

What pattern of deflection of the α -particles was observed?

- A** A few α -particles were deflected through angles greater than a right angle.
- B** All α -particles were deflected from their original path.
- C** Most α -particles were deflected through angles greater than a right angle.
- D** No α -particle was deflected through an angle greater than a right angle.

6 9702/1/O/N/02/Q40

When a nucleus of $^{238}_{92}\text{U}$ absorbs a slow neutron it subsequently emits two β -particles. What is the resulting nucleus?

- A** $^{240}_{93}\text{Np}$ **B** $^{240}_{91}\text{Pa}$ **C** $^{239}_{94}\text{Pu}$ **D** $^{239}_{90}\text{Th}$

7 9702/01/M/J/03/Q38

In what way do the atoms of the isotopes $^{12}_6\text{C}$, $^{13}_6\text{C}$ and $^{14}_6\text{C}$ differ?

- A** different charge **C** different numbers of neutrons
- B** different numbers of electrons **D** different numbers of protons

8 9702/01/M/J/03/Q39

Strontium- 90 ($^{90}_{38}\text{Sr}$) is radioactive and emits β -particles.

Which equation could represent this nuclear decay?

- A** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{39}\text{Sr} + ^0_{-1}\beta$ **C** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{37}\text{Rb} + ^0_1\beta$
- B** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{39}\text{Y} + ^0_{-1}\beta$ **D** $^{90}_{38}\text{Sr} \rightarrow ^{90}_{37}\text{Sr} + ^0_1\beta$

9 9702/01/M/J/03/Q40

Protons and neutrons are thought to consist of smaller particles called quarks.

The 'up' quark has a charge of $\frac{2}{3}e$: a 'down' quark has a charge of $-\frac{1}{3}e$, where e is the elementary charge ($+1.6 \times 10^{-19}\text{C}$).

How many up quarks and down quarks must a proton contain?

	up quarks	down quarks
A	0	3
B	1	1
C	1	2
D	2	1

10 9702/01/O/N/03/Q38

Which are the correct descriptions of a γ -ray and a β -particle?

	γ -ray	β -particle
A	high-speed electron	electromagnetic radiation
B	electromagnetic radiation	Helium-4 nucleus
C	electromagnetic radiation	high-speed electron
D	high-speed electron	Helium-4 nucleus

11 9702/01/O/N/03/Q39

A certain nuclide, Uranium-235, has nucleon number 235, proton number 92 and neutron number 143. Data on four other nuclides are given below.

Which is an isotope of Uranium-235?

	nucleon number	proton number	neutron number
A	235	91	144
B	236	92	144
C	237	94	143
D	238	95	143

12 9702/01/O/N/03/Q40

A nickel nucleus $^{59}_{28}\text{Ni}$ can be transformed by a process termed K-capture. In this process the nucleus absorbs an orbital electron.

If no other process is involved, what is the resulting nucleus?

- A $^{58}_{28}\text{Ni}$ B $^{58}_{27}\text{Co}$ C $^{59}_{27}\text{Co}$ D $^{59}_{29}\text{Cu}$

13 9702/01/M/J/04/Q38

A nucleus of the nuclide $^{241}_{94}\text{Pu}$ decays by emission of a β -particle followed by the emission of an α -particle. Which of the nuclides shown is formed?

- A $^{239}_{93}\text{Np}$ B $^{239}_{91}\text{Pa}$ C $^{237}_{93}\text{Np}$ D $^{237}_{92}\text{U}$

14 9702/01/M/J/04/Q40

Isotopes of a given element all have the same

- A charge/mass ratio. C nucleon number.
B neutron number. D proton number.

15 9702/01/O/N/04/Q38

The symbol $^{77}_{32}\text{Ge}$ represents a nuclide of germanium that decays to a nuclide of arsenic (As) by emitting a β -particle.

What is the symbol of this arsenic nuclide?

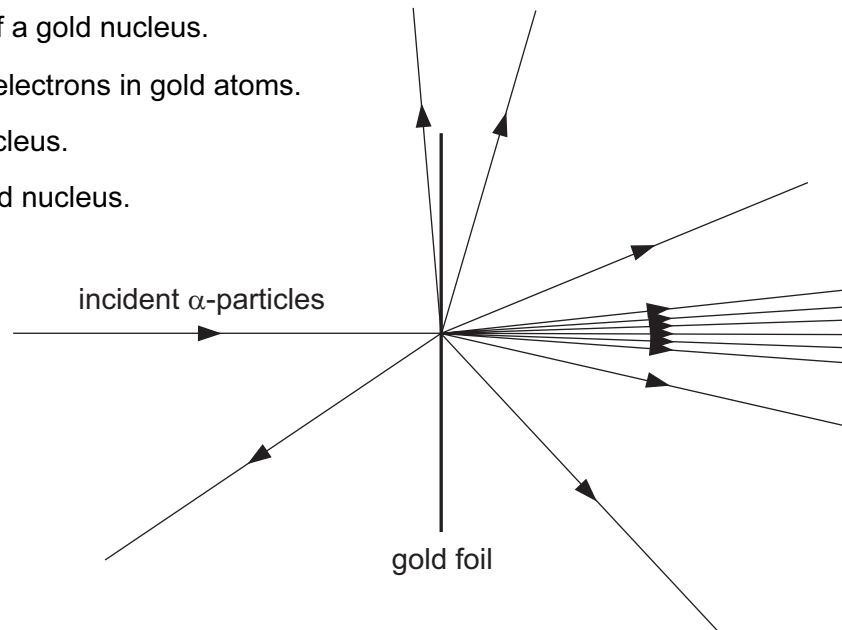
- A $^{76}_{32}\text{As}$ B $^{78}_{32}\text{As}$ C $^{78}_{31}\text{As}$ D $^{77}_{33}\text{As}$

16 9702/01/M/J/04/Q39*

A thin gold foil is bombarded with α -particles as shown.

The results of this experiment provide information about the

- A binding energy of a gold nucleus.
- B energy levels of electrons in gold atoms.
- C size of a gold nucleus.
- D structure of a gold nucleus.



17 9702/01/O/N/04/Q39

The table shows three properties of different types of ionising radiation.

	X	Y	Z
charge	0	$-1e$	$+2e$
mass	0	$\frac{1}{1840}u$	$4u$
speed	c	$\sim 0.9c$	$\sim 0.1c$

What are the radiations X, Y and Z?

	X	Y	Z
A	alpha	beta	X-rays
B	gamma	alpha	beta
C	gamma	beta	alpha
D	X-rays	alpha	beta

18 9702/01/O/N/08/Q38

Which conclusion can be drawn from the results of the experiment showing the scattering of α -particles by gold foil?

- A Electrons orbit the atomic nucleus in well-defined paths.
- B Nuclei of different isotopes contain different numbers of neutrons.
- C The atomic nucleus contains protons and neutrons.
- D The nucleus is very small compared with the size of the atom.

19 9702/01/M/J/09/Q28

Which two nuclei contain the same number of neutrons?

- A** $^{12}_6\text{C}$ and $^{14}_6\text{C}$ **C** $^{23}_{11}\text{Na}$ and $^{24}_{12}\text{Mg}$
B $^{16}_7\text{N}$ and $^{15}_8\text{O}$ **D** $^{32}_{14}\text{Si}$ and $^{32}_{15}\text{P}$

20 9702/01/M/J/05/Q39

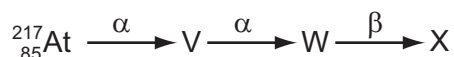
A student conducts an experiment using an α -particle source.

When considering safety precautions, what can be assumed to be the maximum range of α -particles in air?

- A** between 0 and 5 mm **C** between 200 mm and 500 mm
B between 5 mm and 200 mm **D** between 500 mm and 1000 mm

21 9702/01/M/J/05/Q40*

The following represents a sequence of radioactive decays involving two α -particles and one β -particle.



What is the nuclide X?

- A** $^{213}_{85}\text{At}$ **B** $^{215}_{77}\text{Ir}$ **C** $^{209}_{82}\text{Pb}$ **D** $^{217}_{81}\text{Tl}$

22 9702/01/O/N/05/Q38*

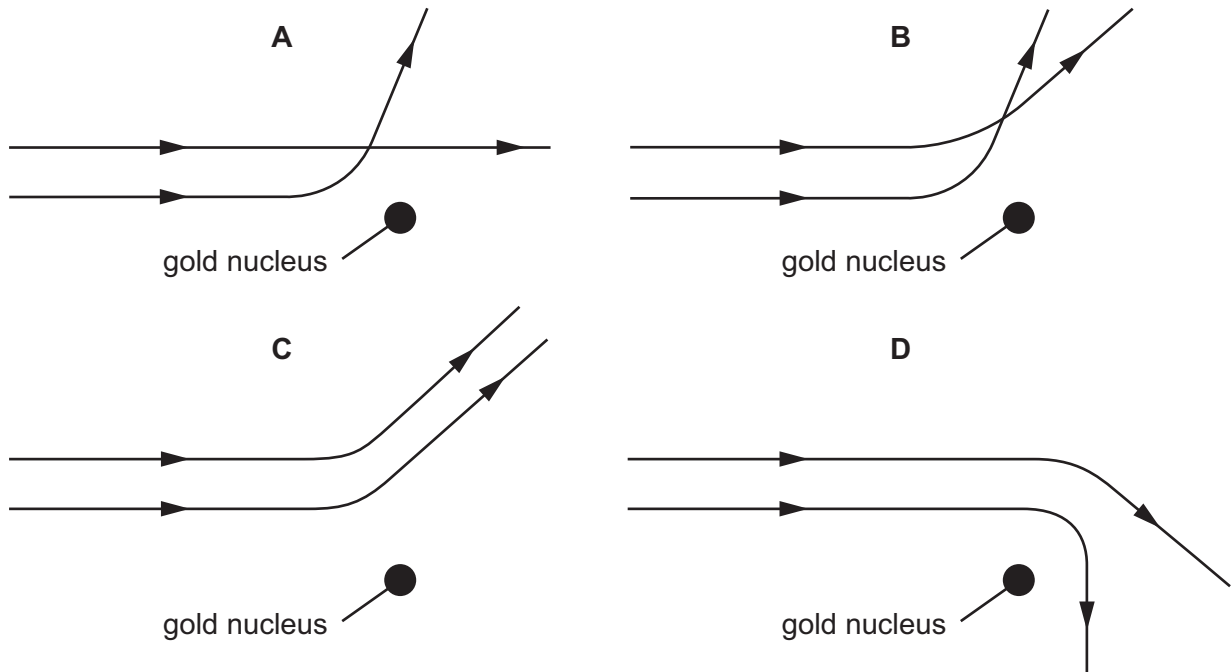
An atomic nucleus emits a β -particle.

What change does this cause to the proton and nucleon numbers of the nucleus?

	proton number	nucleon number
A	-1	+1
B	0	-1
C	+1	-1
D	+1	0

23 9702/01/O/N/05/Q39*

Two α -particles with equal energies are fired towards the nucleus of a gold atom. Which diagram best represents their paths?



24 9702/01/O/N/05/Q40

A nuclear reaction is represented by the equation ${}^{16}_8\text{O} + {}^4_2\text{He} \rightarrow {}^{19}_9\text{F} + \text{X}$. What is particle X?

- A an α -particle B a β -particle C a neutron D a proton

25 9702/01/M/J/06/Q38

What is a correct order of magnitude estimate for the diameter of a typical atomic nucleus?

- A 10^{-14} m B 10^{-18} m C 10^{-22} m D 10^{-26} m

26 9702/01/M/J/06/Q39

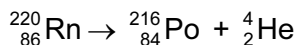
The decay of a nucleus of neptunium is accompanied by the emission of a β -particle and γ -radiation.

What effect (if any) does this decay have on the proton number and the nucleon number of the nucleus?

	proton number	nucleon number
A	increases	decreases
B	decreases	increases
C	unchanged	decreases
D	increases	unchanged

27 9702/01/M/J/06/Q40

Radon-220 is radioactive and decays to Polonium-216 with the emission of an α -particle. The equation for the radioactive decay is shown.



How many neutrons are in the radon and polonium nuclei?

	Rn	Po
A	86	84
B	134	132
C	220	212
D	220	216

28 9702/01/O/N/06/Q38*

Which statement concerning α -particles is correct?

- A An α -particle has charge $+4e$.
- B An α -particle is a helium atom.
- C When α -particles travel through air, they cause ionisation.
- D When α -particles travel through a sheet of gold foil, they make the gold radioactive.

29 9702/01/O/N/06/Q39

Where are electrons, neutrons and protons found in an atom?

	electrons	neutrons	protons
A	in the nucleus	in the nucleus	orbiting the nucleus
B	in the nucleus	orbiting the nucleus	in the nucleus
C	orbiting the nucleus	in the nucleus	orbiting the nucleus
D	orbiting the nucleus	in the nucleus	in the nucleus

9702/01/O/N/06/Q40

30

Radon ${}_{86}^{222}\text{Rn}$ decays by α - and β -emission to bismuth ${}_{83}^{214}\text{Bi}$.

For the decay of each nucleus of radon, how many α - and β -particles are emitted?

	α -particles	β -particles
A	1	1
B	2	1
C	1	2
D	2	2

31 9702/01/M/J/07/Q38

A detector is exposed to a radioactive source. Fluctuations in the count-rate are observed. What do these fluctuations indicate about radioactive decay?

- A It is random.
- B It is spontaneous.
- C It is exponential.
- D It is non-linear.

32 9702/01/M/J/07/Q39

The symbol ${}_{32}^{77}\text{Ge}$ represents a nucleus of germanium that decays to a nucleus of arsenic by emitting a β -particle.

What is the symbol of this arsenic nucleus?

- A ${}_{32}^{76}\text{As}$ B ${}_{32}^{78}\text{As}$ C ${}_{31}^{78}\text{As}$ D ${}_{33}^{77}\text{As}$

33 9702/01/M/J/07/Q40

Each of the nuclei below is accelerated from rest through the same potential difference.

Which one completes the acceleration with the **lowest** speed?

- A ${}^1_1\text{H}$ B ${}^4_2\text{He}$ C ${}^7_3\text{Li}$ D ${}^9_4\text{Be}$

34 9702/01/O/N/07/Q36

How is it possible to distinguish between the isotopes of uranium?

- A Their nuclei have different charge and different mass, and they emit different particles when they decay.
 B Their nuclei have different charge but the same mass.
 C Their nuclei have the same charge but different mass.
 D Their nuclei have the same charge and mass, but they emit different particles when they decay.

35 9702/01/O/N/07/Q37*

What is **not** conserved in nuclear processes?

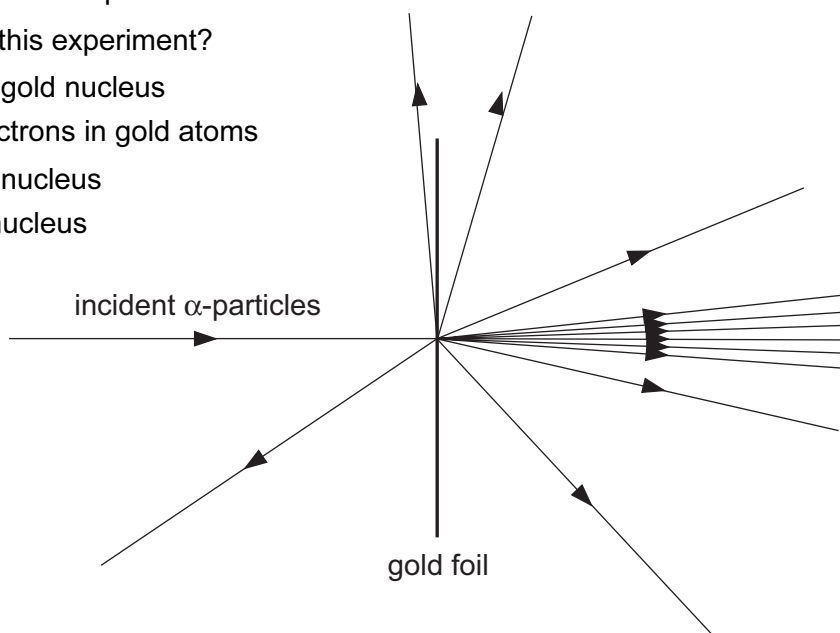
- A energy and mass together C neutron number
 B nucleon number D charge

36 9702/01/O/N/07/Q38

A thin gold foil is bombarded with α -particles as shown.

What can be deduced from this experiment?

- A the binding energy of a gold nucleus
 B the energy levels of electrons in gold atoms
 C the small size of a gold nucleus
 D the structure of a gold nucleus



37 9702/01/O/N/07/Q39

A zirconium nucleus, $^{100}_{40}\text{Zr}$, is a β -emitter. The product nucleus is also a β -emitter.

What is the final resulting nucleus of these two decays?

A $^{100}_{38}\text{Sr}$

B $^{100}_{42}\text{Mo}$

C $^{98}_{40}\text{Zr}$

D $^{102}_{40}\text{Zr}$

38 9702/01/O/N/07/Q40

The following particles are each accelerated from rest through the same potential difference.

Which one completes the acceleration with the **greatest** momentum?

A α -particle

B electron

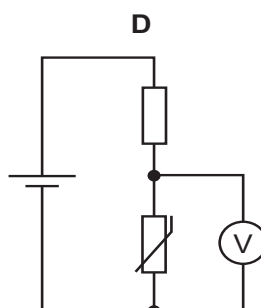
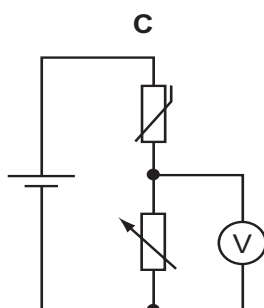
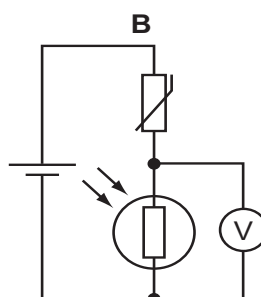
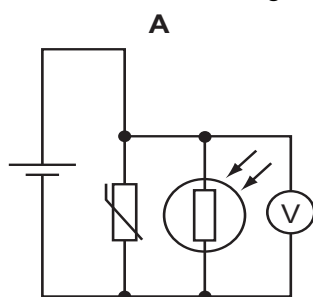
C neutron

D proton

39 9702/01/M/J/08/Q36

A thermistor and another component are connected to a constant voltage supply. A voltmeter is connected across one of the components. The temperature of the thermistor is then reduced but no other changes are made.

In which circuit will the voltmeter reading increase?



40 9702/01/O/N/08/Q38

Which conclusion can be drawn from the results of the experiment showing the scattering of α -particles by gold foil?

A Electrons orbit the atomic nucleus in well-defined paths.

B Nuclei of different isotopes contain different numbers of neutrons.

C The atomic nucleus contains protons and neutrons.

D The nucleus is very small compared with the size of the atom.

41 9702/01/O/N/08/Q39

A nucleus Q has the notation ^y_xQ .

Which of the following is an isotope of Q?

A $^{y-1}_x\text{Q}$

B $^y_{x-1}\text{Q}$

C $^y_{x+1}\text{Q}$

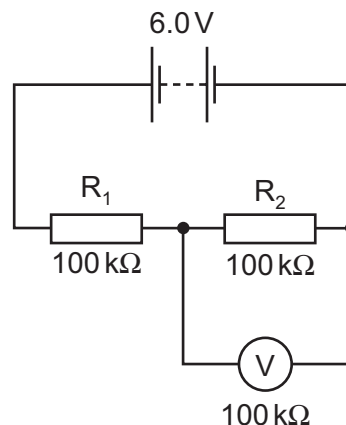
D $^{y-1}_{x+1}\text{Q}$

42 9702/01/M/J/08/Q37

In the circuit shown, the 6.0 V battery has negligible internal resistance. Resistors R_1 and R_2 and the voltmeter have resistance $100\text{ k}\Omega$.

What is the current in the resistor R_2 ?

- A $20\text{ }\mu\text{A}$
- B $30\text{ }\mu\text{A}$
- C $40\text{ }\mu\text{A}$
- D $60\text{ }\mu\text{A}$



43 9702/01/M/J/08/Q39

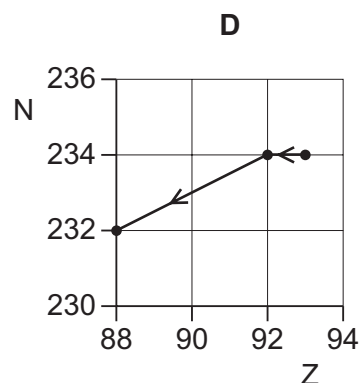
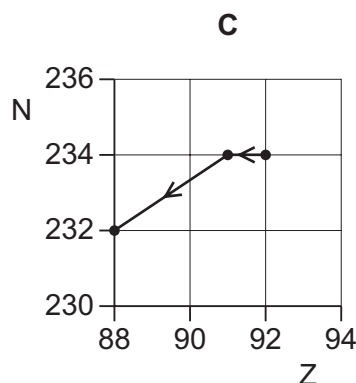
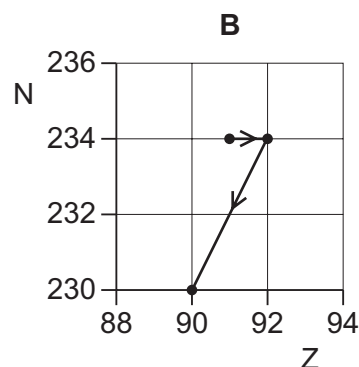
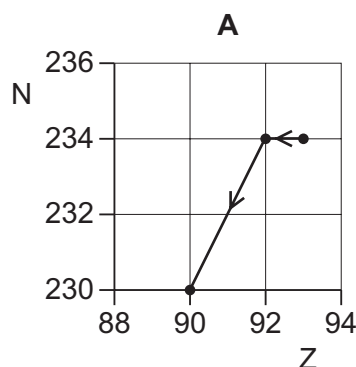
What is the approximate mass of a nucleus of uranium?

- A 10^{-15} kg
- B 10^{-20} kg
- C 10^{-25} kg
- D 10^{-30} kg

44 9702/01/M/J/08/Q40**

A radioactive nucleus is formed by β -decay. This nucleus then decays by α -emission.

Which graph of proton number Z plotted against nucleon number N shows the β -decay followed by the α -emission?



45 9702/01/O/N/08/Q40

A ${}^{238}_{92}\text{U}$ nucleus decays in two stages to a ${}^{234}_{91}\text{Pa}$ nucleus.

What was emitted in these two stages?

- A $\alpha + \beta$
- B $\alpha + \gamma$
- C $\beta + \beta$
- D $\beta + \gamma$

46 9702/01/M/J/09/Q36

How do the nucleon (mass) number and proton (atomic) number of two isotopes of an element compare?

	nucleon number	proton number
A	different	different
B	different	same
C	same	different
D	same	same

47 9702/01/M/J/09/Q37

Nuclear decay is both spontaneous and random.

When the count rate of a radioactive isotope is measured, the readings fluctuate.

Which row describes what the fluctuations demonstrate?

	spontaneous nature	random nature
A	no	no
B	no	yes
C	yes	no
D	yes	yes

48 9702/01/M/J/09/Q38

Which two nuclei contain the same number of neutrons?

- A** $^{12}_6\text{C}$ and $^{14}_6\text{C}$ **C** $^{23}_{11}\text{Na}$ and $^{24}_{12}\text{Mg}$ **B** $^{16}_7\text{N}$ and $^{15}_8\text{O}$ **D** $^{32}_{14}\text{Si}$ and $^{32}_{15}\text{P}$

49 9702/01/M/J/09/Q39

The calcium nuclide $^{42}_{20}\text{Ca}$ is formed by beta decay.

What are the nucleon (mass) number and proton (atomic) number of the unstable nuclide that underwent beta decay to form the calcium nuclide?

	nucleon number	proton number
A	41	19
B	41	21
C	42	19
D	42	21

50 9702/01/M/J/09/Q40

When boron-11 ($^{11}_5\text{B}$) is bombarded with α -particles, a new nucleus is formed and a neutron is released.

Which nuclear equation could represent this reaction?

- A** $^{11}_5\text{B} + ^1_1\text{He} \rightarrow ^{11}_6\text{C} + ^1_0\text{n}$ **C** $^{11}_5\text{B} + ^4_2\text{He} \rightarrow ^{14}_6\text{C} + ^1_1\text{n}$
B $^{11}_5\text{B} + ^2_2\text{He} \rightarrow ^{12}_7\text{N} + ^1_0\text{n}$ **D** $^{11}_5\text{B} + ^4_2\text{He} \rightarrow ^{14}_7\text{N} + ^1_0\text{n}$

51 9702/12/O/N/09/Q36

The gold nucleus $^{185}_{79}\text{Au}$ undergoes alpha decay.

What are the nucleon (mass) number and proton (atomic) number of the nucleus formed by this decay?

	nucleon number	proton number
A	183	79
B	183	77
C	181	77
D	181	75

52 9702/12/O/N/09/Q37

The nuclei of the isotopes of an element all contain the same number of a certain particle.

What is this particle?

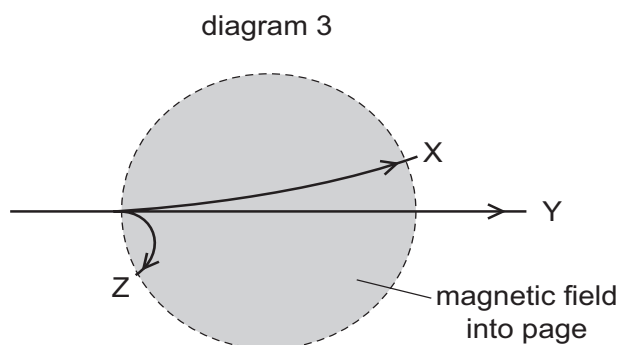
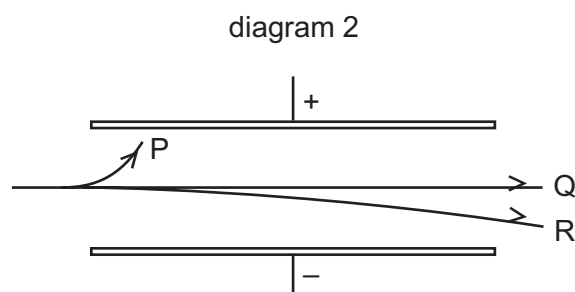
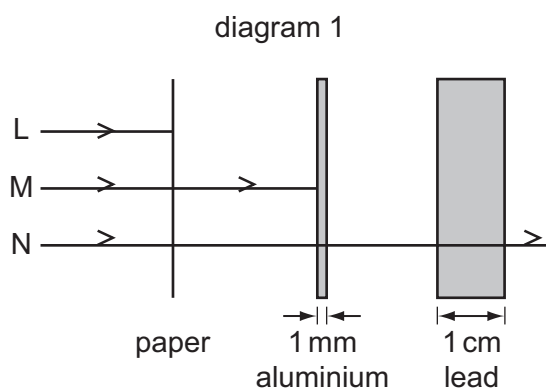
- A electron B neutron C nucleon D proton

53 9702/13/O/N/14/Q40

Alpha, beta and gamma radiations

- are absorbed to different extents in solids,
- behave differently in an electric field,
- behave differently in a magnetic field.

The diagrams illustrate these behaviours.

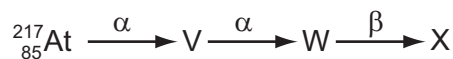


Which three labels on these diagrams refer to the **same** kind of radiation?

- A L, P, X B L, P, Z C M, P, Z D N, Q, X

56 9702/12/M/J/10/Q38

The following represents a sequence of radioactive decays involving two α -particles and one β -particle.



What is the nuclide X?

- A ${}_{85}^{213}\text{At}$ B ${}_{77}^{215}\text{Ir}$ C ${}_{82}^{209}\text{Pb}$ D ${}_{81}^{217}\text{Tl}$

57 9702/12/M/J/10/Q39

What are the correct descriptions of a γ -ray and a β -particle?

	γ -ray	β -particle
A	high-speed electron	electromagnetic radiation
B	electromagnetic radiation	helium-4 nucleus
C	electromagnetic radiation	high-speed electron
D	high-speed electron	helium-4 nucleus

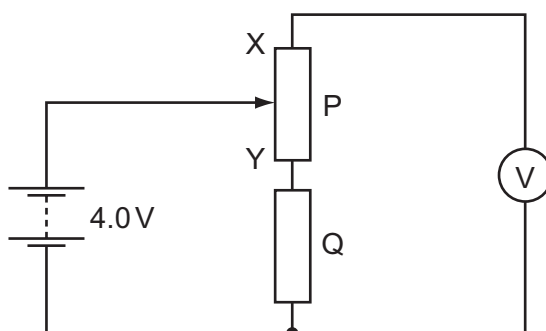
58 9702/12/M/J/10/Q40

What is **not** conserved in nuclear processes?

- A charge C the total number of neutrons
B momentum D the total number of nucleons

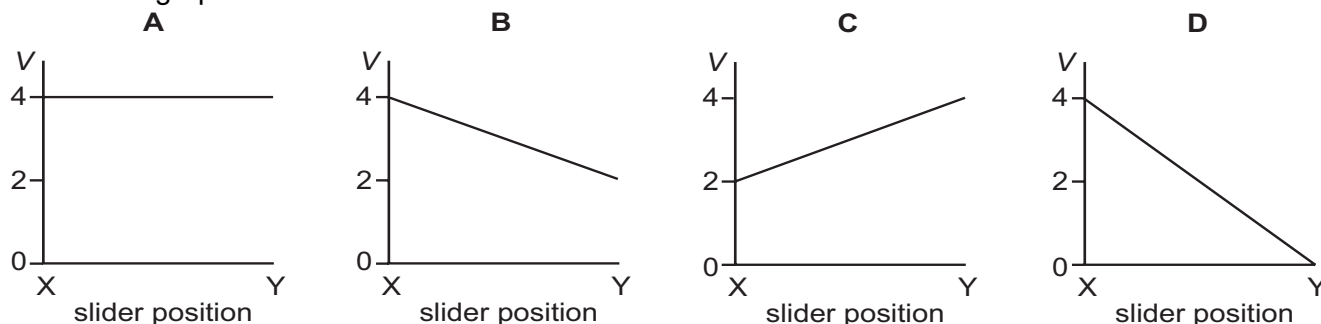
59 9702/11/O/N/10/Q37

In the circuit below, P is a potentiometer of total resistance $10\ \Omega$ and Q is a fixed resistor of resistance $10\ \Omega$. The battery has an e.m.f. of $4.0\ \text{V}$ and negligible internal resistance. The voltmeter has a very high resistance.



The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading V is plotted against slider position.

Which graph would be obtained?



60 9702/11/O/N/10/Q38

Uranium-235 may be represented by the symbol ${}^{235}_{92}\text{U}$.

Which row shows the numbers of nucleons, protons and neutrons in a ${}^{235}_{92}\text{U}$ nucleus?

	nucleons	protons	neutrons
A	92	235	143
B	143	92	235
C	235	92	143
D	235	143	92

61 9702/11/O/N/10/Q39

When a magnesium nucleus ${}^{25}_{12}\text{Mg}$ is hit by a gamma ray, a sodium nucleus ${}^{24}_{11}\text{Na}$ is formed and another particle is emitted.

What are the nucleon number (mass number) and proton number (atomic number) of the other particle produced in this nuclear reaction?

	nucleon number	proton number
A	0	-1
B	0	1
C	1	-1
D	1	1

62 9702/11/O/N/10/Q40

Which nuclear equation shows the beta decay of a nucleus of argon (Ar) into potassium (K)?



63 9702/12/O/N/10/Q37

Three resistors, with resistances R_1 , R_2 and R_3 , are connected in series and are found to have a combined resistance of $500\ \Omega$. When connected in parallel, the combined resistance is found to be $50\ \Omega$.

Which values will correspond to these results?

	R_1/Ω	R_2/Ω	R_3/Ω
A	160	160	80
B	200	200	100
C	225	225	50
D	230	230	40

64 9702/12/O/N/10/Q38

In the Rutherford scattering experiment, α -particles were fired at a thin gold foil. A small proportion of the α -particles were deflected through large angles.

Which statement gives the correct conclusion that could be drawn directly from these results?

- A** The atom is made up of electrons, protons and neutrons.
- B** The nucleus is at the centre of the atom.
- C** The nucleus is made up of protons and neutrons.
- D** The atom contains a very small, charged nucleus.

65 9702/12/O/N/10/Q39

Which statement about the nuclei of the atoms of an element is correct?

- A** Every nucleus of an element contains an equal number of neutrons and protons.
- B** Every nucleus of an element contains the same number of neutrons as all others of that element, but the number of protons may differ.
- C** Every nucleus of an element contains the same number of protons as all others of that element, but the number of neutrons may differ.
- D** The number of protons in a nucleus differs from isotope to isotope of an element, as do the number of neutrons.

66 9702/12/O/N/10/Q40

A counter recording radioactive decays from a radioactive source gives the following counts in equal intervals of time.

time / min	counts
0–10	424
10–20	395
20–30	413
30–40	363
40–50	366
50–60	294
60–70	301
70–80	253
80–90	212

What can be deduced from these readings?

- A** that radioactivity is random and that the half-life is 90 minutes
- B** that radioactivity is random and that the half-life is uncertain
- C** that radioactivity is spontaneous and that the half-life is 90 minutes
- D** that radioactivity is spontaneous and that the half-life is uncertain

67 9702/11/M/J/11/Q38

The first artificial radioactive substance was made by bombarding aluminium, $^{27}_{13}\text{Al}$, with α -particles. This produced an unstable isotope of phosphorus, $^{30}_{15}\text{P}$.

What was the by-product of this reaction?

- A** an α -particle **B** a β -particle **C** a γ -ray **D** a neutron

68 9702/11/M/J/11/Q39

Uranium-238, $^{238}_{92}\text{U}$, decays by α -emission into a daughter product which in turn decays by β -emission into a grand-daughter product.

What is the grand-daughter product?

- A** $^{234}_{90}\text{Th}$ **B** $^{234}_{91}\text{Pa}$ **C** $^{234}_{92}\text{U}$ **D** $^{230}_{90}\text{Th}$

69 9702/11/M/J/11/Q40

Which statement about nuclei is correct?

- A** Different isotopic nuclei have different proton numbers.
B For some nuclei, the nucleon number can be less than the proton number.
C In some nuclear processes, mass-energy is not conserved.
D Nucleon numbers of nuclei are unchanged by the emission of β -particles.

70 9702/12/M/J/11/Q39

The uranium nucleus $^{238}_{92}\text{U}$ undergoes α -decay, producing nucleus X.

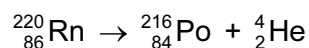
Nucleus X undergoes β -decay, producing nucleus Y.

For nucleus Y, what are the values of the proton number and nucleon number?

	proton number	nucleon number
A	89	234
B	89	236
C	91	234
D	91	236

71 9702/12/M/J/11/Q40

Radon-220 is radioactive and decays to polonium-216 with the emission of an α -particle. The equation for the radioactive decay is shown.



How many neutrons are in the radon and polonium nuclei?

	Rn	Po
A	86	84
B	134	132
C	220	212
D	220	216

72 9702/11/O/N/11/Q40

An atomic nucleus emits a β -particle.

What change does this cause to the proton number and the nucleon number of the nucleus?

	proton number	nucleon number
A	-1	+1
B	0	-1
C	+1	-1
D	+1	0

73 9702/12/O/N/11/Q39

Which statement concerning α -particles is correct?

- A** An α -particle has charge $+4e$.
- B** An α -particle is a helium atom.
- C** When α -particles travel through air, they cause ionisation.
- D** When α -particles travel through a sheet of gold foil, they make the gold radioactive.

74 9702/12/O/N/11/Q40

A nucleus of the nuclide ${}_{94}^{241}\text{Pu}$ decays by emission of a β -particle followed by the emission of an α -particle.

Which nucleus is formed?

- A** ${}_{93}^{239}\text{Np}$ **B** ${}_{91}^{239}\text{Pa}$ **C** ${}_{93}^{237}\text{Np}$ **D** ${}_{92}^{237}\text{U}$

75 9702/12/M/J/12/Q39

Nuclear decay is both spontaneous and random in nature.

Which row gives the correct experimental evidence for these properties?

	spontaneous nature of decay	random nature of decay
A	the decay rate is not affected by pressure	the decay rate is not affected by temperature
B	the decay rate is not affected by pressure	the rate at which radiation is received at a counter fluctuates
C	the decay rate is not affected by temperature	the decay rate is not affected by pressure
D	the rate at which radiation is received at a counter fluctuates	the decay rate is not affected by pressure

76 9702/12/M/J/12/Q40

Radon $^{222}_{86}\text{Rn}$ is the start of a decay chain that forms bismuth $^{214}_{83}\text{Bi}$ by alpha and beta emission.

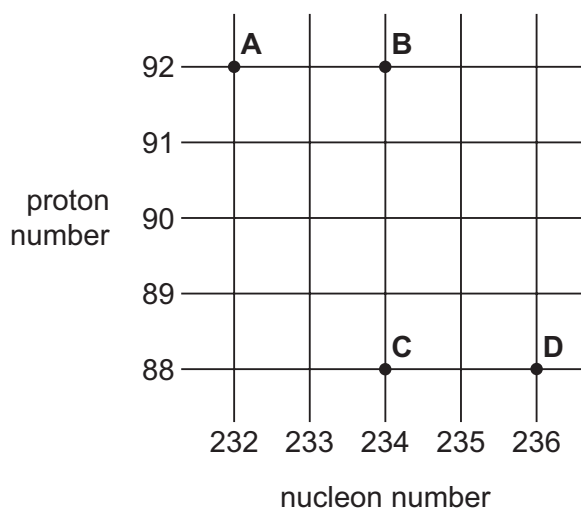
For the decay of each nucleus of radon, how many α -particles and β -particles are emitted?

	α -particles	β -particles
A	1	1
B	2	1
C	1	2
D	2	2

77 9702/13/M/J/12/Q39

Thorium-234 ($^{234}_{90}\text{Th}$) decays by β -emission into a daughter product which in turn decays by further β -emission into a granddaughter product.

Which letter in the diagram represents the granddaughter product?



78 9702/13/M/J/12/Q40

An experiment in which α -particles were deflected by a gold foil produced new insights into the structure of the atom.

Which conclusion can be drawn from the results of the experiment?

- A** Atomic nuclei occupy a very small fraction of the volume of an atom.
- B** Electrons orbit the atomic nucleus.
- C** Some atoms of the same element contain different numbers of neutrons.
- D** The atomic nucleus contains protons and neutrons.

79 9702/12/O/N/12/Q39

A material contains a radioactive isotope that disintegrates solely by the emission of α -particles at a rate of 100 s^{-1} .

Which statement about this material is correct?

- A** The number of atoms in the material diminishes at a rate of 100 s^{-1} .
- B** The number of neutrons in the material diminishes at a rate of 100 s^{-1} .
- C** The number of nucleons in the material diminishes at a rate of 400 s^{-1} .
- D** The number of protons in the material diminishes at a rate of 100 s^{-1} .

80 9702/12/O/N/12/Q40

A different nucleus can be formed by bombarding a stable nucleus with an energetic α -particle.

$^{23}_{11}\text{Na}$ is bombarded with an energetic α -particle.

What could be the products of this nuclear reaction?

- A** $^{25}_{10}\text{Ne}$ + neutron **B** $^{25}_{11}\text{Na}$ + proton **C** $^{26}_{12}\text{Mg}$ + β **D** $^{27}_{13}\text{Al}$ + γ

81 9702/13/O/N/12/Q38

A nuclear isotope emits radiation which is detected by a Geiger-Müller tube held at a distance of about 10 cm from the radioactive source. The radiation is stopped completely by a 2 mm thick sheet of lead.

What can be deduced from this information about the emission from the isotope?

- A** It could be alpha and beta radiation, but not gamma radiation.
- B** It could be alpha and gamma radiation, but not beta radiation.
- C** It could be beta and gamma radiation, but not alpha radiation.
- D** It could be alpha, beta and gamma radiation.

82 9702/13/O/N/12/Q39

What remains constant during β -emission from a number of identical nuclei in a substance?

- A** energy of the β -particles **C** nucleon number of the nuclei
- B** neutron number of the nuclei **D** proton number of the nuclei

83 9702/11/M/J/13/Q38

A nickel nucleus $^{59}_{28}\text{Ni}$ can be transformed by a process termed K-capture. In this process the nucleus absorbs an orbital electron.

If no other process is involved, what is the resulting nucleus?

- A** $^{58}_{28}\text{Ni}$ **B** $^{58}_{27}\text{Co}$ **C** $^{59}_{27}\text{Co}$ **D** $^{59}_{29}\text{Cu}$

84 9702/11/M/J/13/Q39

It was once thought that the mass of an atom is spread uniformly through the volume of the atom. When α -particles are directed at a piece of gold foil, the results led scientists to believe instead that nearly all the mass of the gold atom is concentrated at a point inside the atom.

Which effect is possible **only** if nearly all the mass of the gold atom is concentrated at a point?

- A** a few α -particles bounce back
B most α -particles are only slightly deflected
C some α -particles pass through without any deflection
D some α -particles are absorbed

85 9702/13/M/J/13/Q38

Scientists investigating the count rate from a radioactive source observed that the count rate fluctuates.

What do these fluctuations imply about the nature of radioactive decay?

- A** It involves atomic nuclei. **B** It is predictable. **C** It is random. **D** It is spontaneous.

86 9702/11/M/J/13/Q40

Which pair of nuclei are isotopes of one another?

	nucleon number	number of neutrons
A	186	112
	180	118
B	186	112
	182	108
C	184	110
	187	110
D	186	110
	186	112

87 9702/12/M/J/13/Q39

What is the approximate mass of an alpha particle?

- A** 10^{-28} kg **B** 10^{-26} kg **C** 10^{-24} kg **D** 10^{-22} kg

88 9702/12/M/J/13/Q40

An actinium nucleus has a nucleon number of 227 and a proton number of 89. It decays to form a radium nucleus, emitting a beta particle and an alpha particle in the process.

What are the nucleon number and the proton number of this radium nucleus?

	nucleon number	proton number
A	223	87
B	223	88
C	224	87
D	225	86

89 9702/13/M/J/13/Q39

The decay of a nucleus of neptunium is accompanied by the emission of a β -particle and γ -radiation.

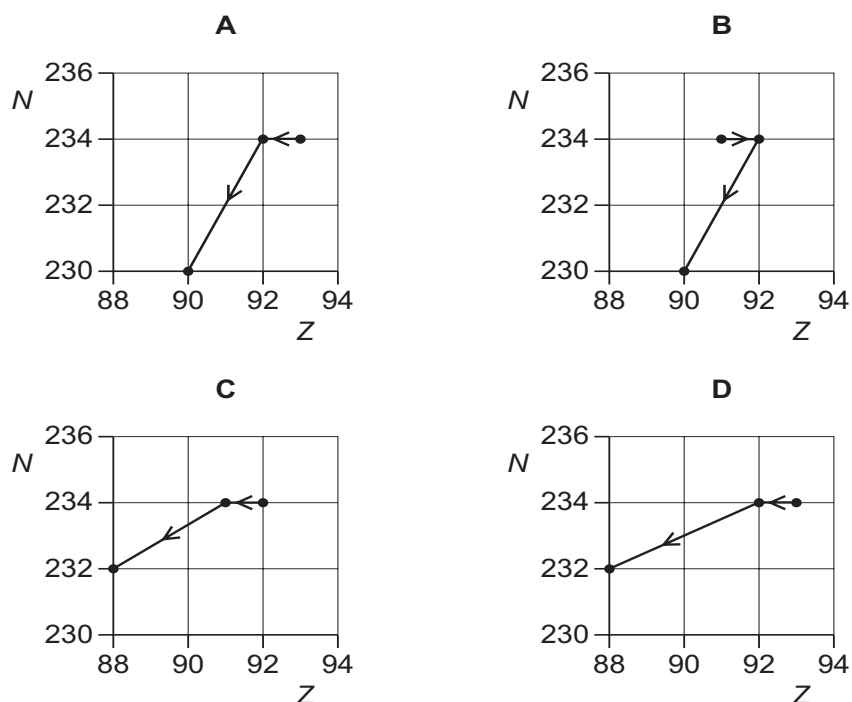
What effect (if any) does this decay have on the proton number and on the nucleon number of the nucleus?

	proton number	nucleon number
A	increases	decreases
B	decreases	increases
C	unchanged	decreases
D	increases	unchanged

90 9702/13/M/J/13/Q40

A radioactive nucleus is formed by β -decay. This nucleus then decays by α -emission.

Which graph of nucleon number N plotted against proton number Z shows the β -decay followed by the α -emission?



91 9702/12/O/N/13/Q39

When α -particles are fired at a thin metal foil, most of the particles pass straight through but a few are deflected by a large angle.

Which change would increase the **proportion** of α -particles deflected by a large angle?

- A using α -particles with greater kinetic energy
- B using a foil made of a metal with fewer protons in its nuclei
- C using a double thickness foil
- D using an alpha source with a higher activity

92 9702/12/O/N/13/Q40

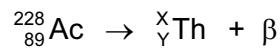
Plutonium-239 ($^{239}_{94}\text{Pu}$) decays by emitting α -radiation.

Which nuclide is formed from one of these decay reactions? (The product nuclides are represented by X.)

- A $^{235}_{92}\text{X}$ B $^{237}_{92}\text{X}$ C $^{239}_{93}\text{X}$ D $^{239}_{95}\text{X}$

93 9702/13/O/N/13/Q39

A nucleus of the nuclide $^{228}_{89}\text{Ac}$ decays by emitting a beta particle. The nuclear equation below represents this decay.



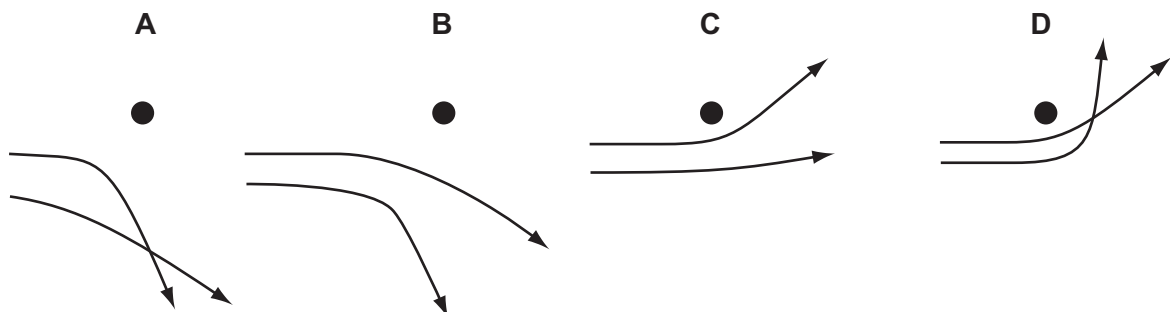
Which pair of values of X and Y is correct?

	X	Y
A	224	87
B	224	89
C	228	88
D	228	90

94 9702/13/O/N/13/Q40

Two α -particles with equal energies are deflected by a large nucleus.

Which diagram best represents their paths?



95 9702/11/M/J/14/Q37

Alpha, beta and gamma radiations have various depths of penetration in matter and different charges.

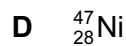
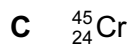
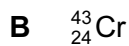
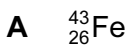
Which row best summarises the penetration and charge of each radiation?

	alpha	beta	gamma
A	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium no charge	not fully absorbed by several cm of lead no charge
B	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium positive charge	not fully absorbed by several cm of lead no charge
C	absorbed by a sheet of card positive charge	absorbed by several mm of aluminium negative charge	not fully absorbed by several cm of lead no charge
D	absorbed by several mm of aluminium positive charge	not fully absorbed by several cm of lead negative charge	absorbed by a sheet of card no charge

96 9702/11/M/J/14/Q38

In 2002, two-proton radioactive decay of an isotope of iron, ${}^{45}_{26}\text{Fe}$, was observed.

What could be the resulting product?



97 9702/11/M/J/14/Q39

U^{++} is a doubly-ionised uranium atom. The uranium atom has a nucleon number of 235 and a proton number of 92.

In a simple model of the atom, how many particles are in this ionised atom?

A 235

B 325

C 327

D 329

98 9702/11/M/J/14/Q40

The grid shows a number of nuclides arranged according to the number of protons and the number of neutrons in each.

A nucleus of the nuclide ${}^8_3\text{Li}$ decays by emitting a β -particle.

What is the resulting nuclide?

					A	B	
4							
3				${}^6_3\text{Li}$	${}^7_3\text{Li}$	${}^8_3\text{Li}$	
2		${}^3_2\text{He}$	${}^4_2\text{He}$			C	D
1	${}^1_1\text{H}$	${}^2_1\text{H}$					
	0	1	2	3	4	5	6
	number of neutrons						

99 9702/12/M/J/14/Q38

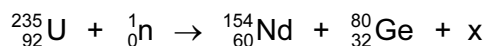
A nucleus X decays into a nucleus Y by emitting an alpha particle followed by two beta particles.

Which statement about this nuclear decay is correct?

- A Beta particle decay occurs when a proton changes into a neutron.
- B Nucleus Y has the same nucleon number as nucleus X.
- C Nucleus Y is an isotope of nucleus X.
- D The total mass of the products is equal to the mass of the initial nucleus X.

100 9702/12/M/J/14/Q39

A slow-moving neutron collides with a nucleus of uranium-235. This results in a nuclear reaction that is represented by the following nuclear equation



where x represents one or more particles.

What does x represent?

- A one neutron
- B two electrons
- C two neutrons
- D two protons

101 9702/12/M/J/14/Q40

The first artificial radioactive substance was made by bombarding aluminium, ${}^{27}_{13}\text{Al}$, with α -particles. This produced an unstable isotope of phosphorus, ${}^{30}_{15}\text{P}$.

What was the by-product of this reaction?

- A an α -particle
- B a β -particle
- C a neutron
- D a proton

102 9702/13/M/J/14/Q39

An isotope of thorium has a nucleon number of 232 and a proton number of 90. It decays to form another isotope of thorium with a nucleon number of 228.

How many alpha particles and beta particles are emitted by a nucleus of thorium during this decay?

	alpha particles	beta particles
A	0	4
B	1	0
C	1	2
D	2	1

103 9702/13/M/J/14/Q40

Four nuclei are represented below.



Which statement about these nuclei is correct?

- A** An uncharged atom of element Q has 24 orbital electrons.
- B** Nucleus M could transform into Q by emitting a beta particle.
- C** Nuclei G and M are isotopes of the same element.
- D** When E absorbs a neutron and then emits an alpha particle, nucleus E transforms into M.

104 9702/11/O/N/14/Q35

Which statement about α -particles is correct?

- A** α -particles emitted from a single radioactive isotope have a continuous distribution of energies.
- B** α -particles have less ionising power than β -particles.
- C** The charge of an α -particle is $+1.60 \times 10^{-19} \text{ C}$.
- D** The speeds of α -particles can be as high as $1.5 \times 10^7 \text{ m s}^{-1}$.

105 9702/11/O/N/14/Q39

The isotope ${}_{86}^{222}\text{Rn}$ decays in a sequence of emissions to form the isotope ${}_{82}^{206}\text{Pb}$. At each stage of the decay sequence, it emits either an α -particle or a β -particle.

What is the number of stages in the decay sequence?

- A** 4
- B** 8
- C** 16
- D** 20

106 9702/12/O/N/14/Q40

What is the approximate mass of a nucleus of uranium?

- A** 10^{-15} kg **B** 10^{-20} kg **C** 10^{-25} kg **D** 10^{-30} kg

107 9702/13/O/N/14/Q38

The nucleus of a radioactive isotope of an element emits an alpha particle. The daughter nucleus then emits a beta particle and then the daughter nucleus of that reaction emits another beta particle.

Which statement describes the final nuclide that is formed?

- A** It is a different isotope of the original element.
B It is a nuclide of a different element of higher proton number.
C It is a nuclide of the same element but with different proton number.
D It is identical to the original nuclide.

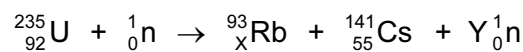
108 9702/13/O/N/14/Q39

A nuclear reaction is shown. ${}^{238}_{92}\text{U} + {}^4_2\text{He} \rightarrow {}^{241}_{94}\text{Pu} + \text{X}$ What is product X?

- A** an alpha particle **B** an electron **C** a neutron **D** a proton

109 9702/13/M/J/15/Q40

The nuclear equation for a fission reaction is shown below.



What are the values of X and Y?

	X	Y
A	37	0
B	37	1
C	37	2
D	38	2

110 9702/11/M/J/15/Q39

When α -particles are directed at gold leaf

- 1 almost all α -particles pass through without deflection,
- 2 a few α -particles are deviated through large angles.

What are the reasons for these effects?

	1	2
A	most α -particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α -particles bounce back from the gold surface
B	most α -particles miss all gold atoms	a few α -particles bounce off gold atoms
C	the gold nucleus is very small so most α -particles miss all nuclei	occasionally the path of an α -particle is close to a nucleus
D	the positive charge in an atom is not concentrated enough to deflect an α -particle	occasionally an α -particle experiences many small deflections in the same direction

111 9702/11/M/J/15/Q40

The nuclide $^{222}_{86}\text{Rn}$ decays in a sequence of stages to form the nuclide $^{206}_{82}\text{Pb}$.

Four of the nuclides formed in the sequence are α -particle emitters. The others are β -particle emitters.

How many nuclides formed in the decay sequence are β -particle emitters?

- A** 2 **B** 4 **C** 8 **D** 12

112 9702/12/M/J/15/Q38

An experiment in which α -particles were deflected by a gold foil produced new insights into the structure of the atom.

Which conclusion can be drawn from the results of the experiment?

- A** Atomic nuclei occupy a very small fraction of the volume of an atom.
B Electrons orbit the atomic nucleus.
C Some atoms of the same element contain different numbers of neutrons.
D The atomic nucleus contains protons and neutrons.

113 9702/12/M/J/15/Q39

A radioactive substance contains a number of identical nuclei that emit β -particles.

Which property of these nuclei remains unaltered by the emission?

- A** charge **B** neutron number **C** nucleon number **D** proton number

114 9702/12/M/J/15/Q40

A uranium-238 nucleus, ${}^{238}_{92}\text{U}$, undergoes nuclear decays to form uranium-234, ${}^{234}_{92}\text{U}$.

Which series of decays could give this result?

- A** emission of four β -particles **C** emission of one α -particle and two β -particles
B emission of four γ -rays **D** emission of two α -particles and eight β -particles

115 9702/13/M/J/15/Q38

Nucleus P decays in two stages to produce nucleus Q.

Which decay sequence will result in the highest number of neutrons in nucleus Q?

- A** an α -particle followed by a β -particle **C** a β -particle followed by another β -particle
B an α -particle followed by a γ -ray **D** a β -particle followed by a γ -ray

116 9702/13/M/J/15/Q39

A neutral atom has a nucleus given by the symbol ${}^{133}_{55}\text{Cs}$.

How many protons, neutrons and electrons are in this atom?

	protons	neutrons	electrons
A	55	78	55
B	55	133	55
C	78	55	78
D	133	55	133

117 9702/12/M/J/17/Q39

A certain nuclide, uranium-235, has nucleon number 235, proton number 92 and neutron number 143. Data on four other nuclides are given below.

Which nuclide is an isotope of uranium-235?

	nucleon number	proton number	neutron number
A	235	91	144
B	236	92	144
C	237	94	143
D	238	95	143

118 9702/12/M/J/17/Q40

During β^- decay, which change takes place to the quark composition of the nucleus that emits the β^- particle, and which other particle is emitted?

	quark change	other particle emitted
A	down to up	antineutrino
B	down to up	neutrino
C	up to down	antineutrino
D	up to down	neutrino

119 9702/11/M/J/17/Q38

The table lists the nucleon number and the proton number of various nuclei. The nuclei are represented by the letters L to T.

nucleus	nucleon number	proton number
L	227	89
M	226	89
N	225	89
O	227	90
P	226	90
Q	225	90
R	227	91
S	226	91
T	225	91

Which row in the following table correctly shows three nuclei of the same element, and three nuclei that have the same number of neutrons?

	same element	same number of neutrons
A	L M N	R P N
B	M P S	R S T
C	O P Q	M P S
D	R P N	O P Q

120 9702/12/M/J/17/Q38

In a television programme to illustrate scientific models, a presenter fires a gun many times at a bale of hay. Two small cannon balls are embedded within the hay some distance apart from each other.

The hay bale measures approximately $2\text{ m} \times 2\text{ m} \times 2\text{ m}$ and the cannon balls are made of iron, approximately spherical, and about 5 cm in diameter.

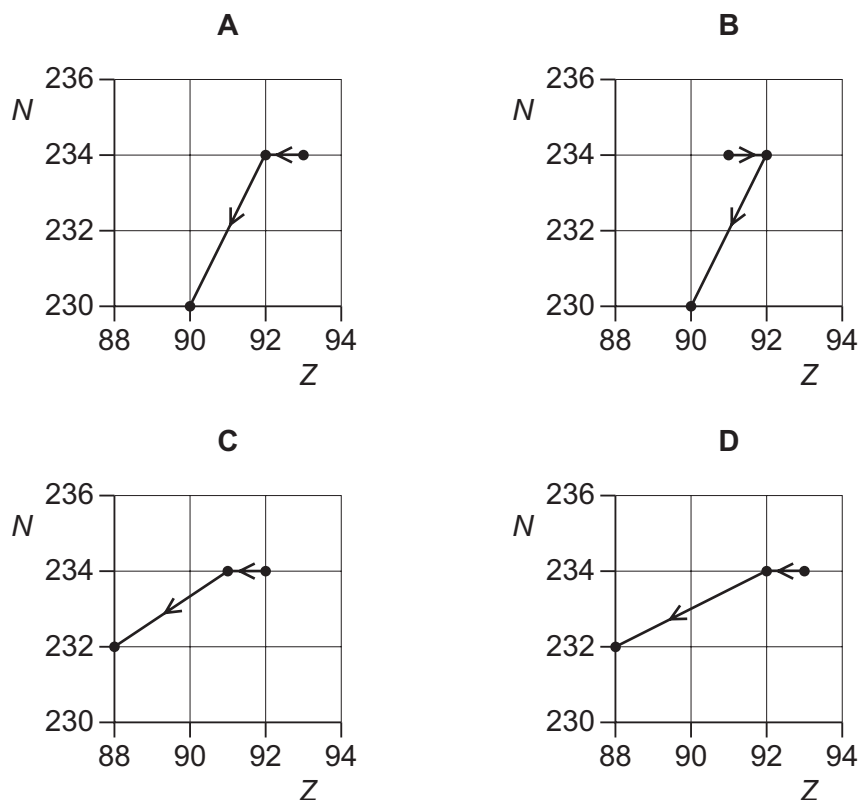
What might the presenter be illustrating?

- A** α -particle scattering
- B** β^- decay
- C** conservation of momentum
- D** double-slit interference

121 9702/01/M/J/08/Q40

A radioactive nucleus is formed by β^- decay. This nucleus then decays by α -emission.

Which graph of nucleon number N plotted against proton number Z shows the β^- decay followed by the α -emission?



122 9702/11/M/J/17/Q40

What are the structures of the proton and of the neutron in terms of quarks?

	proton		neutron	
	up quark	down quark	up quark	down quark
A	1	1	2	2
B	1	2	2	1
C	2	1	1	2
D	2	2	1	1

123 9702/11/O/N/17/Q39

An astatine nucleus has a nucleon number of 218 and a proton number of 85. It decays to form a polonium nucleus, emitting a β^- particle and an α -particle in the process.

What are the nucleon number and the proton number of this polonium nucleus?

	nucleon number	proton number
A	214	83
B	214	84
C	215	83
D	216	82

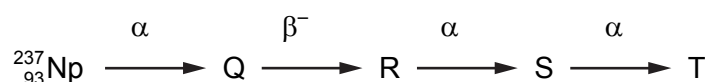
124 9702/11/O/N/17/Q40

What is the quark composition of a hydrogen-3 nucleus, ${}^3_1\text{H}$?

	number of quarks	
	up	down
A	4	5
B	5	4
C	5	7
D	7	5

125 9702/12/O/N/17/Q39

The diagram shows a sequence of radioactive decays involving three α -particles and a β^- particle.



What is nuclide T?

- A** ${}^{225}_{88}\text{Ra}$ **B** ${}^{231}_{88}\text{Ra}$ **C** ${}^{225}_{90}\text{Th}$ **D** ${}^{229}_{90}\text{Th}$

126 9702/12/O/N/17/Q40

Which combination of up (u) and down (d) quarks forms a neutron?

- A** u u u **B** u u d **C** u d d **D** d d d

127 9702/13/O/N/17/Q39

Each option shows the number of neutrons, protons and electrons in two atoms, some of which are ionised.

Which pair of atoms in the table are isotopes of the same element?

	neutron	proton	electron
A	20 20	17 19	20 20
B	28 28	23 24	23 24
C	34 36	29 29	29 28
D	40 42	32 31	32 32

128 9702/13/M/J/18/Q40

Which list contains only leptons?

- A** electron, neutrino, positron **C** electron, proton, neutron
B electron, neutrino, proton **D** neutrino, neutron, positron

129 9702/11/M/J/18/Q38

Which elementary particle is a lepton?

- A** proton **B** neutron **C** electron **D** quark

130 9702/11/M/J/18/Q39

How many down quarks are in a nucleus of hydrogen-3, ${}^3_1\text{H}$?

- A** 2 **B** 3 **C** 4 **D** 5

131 9702/11/M/J/18/Q40

What is the correct equation for β^+ decay?

- A** neutron $\rightarrow$ proton + electron + electron antineutrino
B neutron $\rightarrow$ proton + electron + electron neutrino
C proton $\rightarrow$ neutron + positron + electron antineutrino
D proton $\rightarrow$ neutron + positron + electron neutrino

132 9702/12/M/J/18/Q38

In the α -particle scattering experiment, a beam of α -particles is aimed at a thin gold foil. Most of the α -particles go straight through or are deflected by a small angle. A very small proportion are deflected through more than 90° , effectively rebounding towards the source of the α -particles.

Which conclusion about the structure of atoms **cannot** be drawn from this experiment alone?

- A** Most of the atom is empty space.
B Most of the mass of an atom is concentrated in the nucleus.
C The nucleus contains both protons and neutrons.
D The nucleus is charged.

133 9702/12/M/J/18/Q39

Radon-211, $^{211}_{86}\text{Rn}$, francium-210, $^{210}_{87}\text{Fr}$, and radium-212, $^{212}_{88}\text{Ra}$, are three nuclides.

How many neutrons does each nuclide have in its nucleus?

	radon-211	francium-210	radium-212
A	86	87	88
B	125	123	124
C	211	210	212
D	297	297	300

134 9702/12/M/J/18/Q40

A neutron is composed of one up (u) quark and two down (d) quarks. When the neutron decays to a proton, there is β -emission.

What is the change in the quark structure of the neutron due to the β -emission?

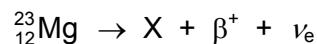
(The symbol for a neutrino is ν_e and for an antineutrino is $\bar{\nu}_e$.)

A $d \rightarrow u + \beta^- + \nu_e$ **C** $u \rightarrow d + \beta^+ + \nu_e$

B $d \rightarrow u + \beta^- + \bar{\nu}_e$ **D** $u \rightarrow d + \beta^+ + \bar{\nu}_e$

135 9702/13/M/J/18/Q39

A nucleus of magnesium-23 undergoes β^+ decay, as represented by the nuclear equation shown.



What is nucleus X?

A $^{22}_{11}\text{Na}$

B $^{22}_{13}\text{Al}$

C $^{23}_{11}\text{Na}$

D $^{23}_{13}\text{Al}$

136 9702/12/F/M/18/Q40

An isolated neutron decays to produce a proton, a β^- particle and an antineutrino.

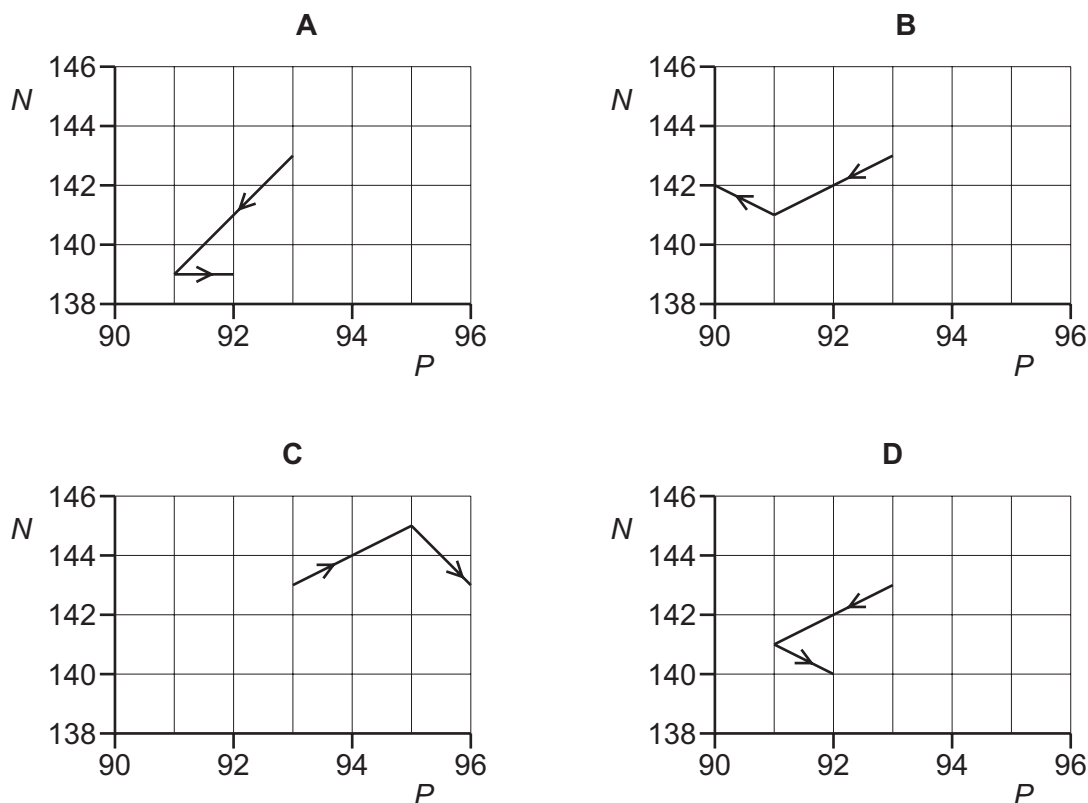
Which row gives the quark composition of the neutron and the proton and the type of force that gives rise to this reaction?

	quark composition		type of force
	neutron	proton	
A	down, down, up	down, up, up	strong interaction
B	down, down, up	down, up, up	weak interaction
C	down, up, up	down, down, up	strong interaction
D	down, up, up	down, down, up	weak interaction

137 9702/12/F/M/18/Q39

A nucleus of neptunium-236 contains 93 protons and 143 neutrons. This nucleus decays with the emission of an α -particle. The nucleus formed then emits a β^- particle.

Which diagram shows the changes in the number P of protons and the number N of neutrons in these nuclei?



138 9702/12/F/M/18/Q40

An isolated neutron decays to produce a proton, a β^- particle and an antineutrino.

Which row gives the quark composition of the neutron and the proton and the type of force that gives rise to this reaction?

	quark composition		type of force
	neutron	proton	
A	down, down, up	down, up, up	strong interaction
B	down, down, up	down, up, up	weak interaction
C	down, up, up	down, down, up	strong interaction
D	down, up, up	down, down, up	weak interaction

139 9702/11/O/N/18/Q39

A nucleus emits a β^- particle.

What is the change to the proton number and to the nucleon number of the nucleus?

	proton number	nucleon number
A	-1	+1
B	0	-1
C	+1	-1
D	+1	0

140 9702/11/O/N/18/Q40

How many up quarks and how many down quarks are in a nucleus of the nuclide ${}^{37}_{17}\text{Cl}$?

	up quarks	down quarks
A	51	60
B	54	57
C	57	54
D	60	51

141 9702/12/O/N/18/Q39

A proton in a nucleus undergoes β^+ decay. One of the products is a neutron.

What are the other products?

- A** an electron and a neutrino
- B** an electron and an antineutrino
- C** a positron and a neutrino
- D** a positron and an antineutrino

142 9702/12/O/N/18/Q40

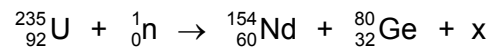
A certain type of hadron has zero charge. It is composed of a down quark, a strange quark and one other quark.

What could be the other quark?

- A** up
- B** down
- C** strange
- D** anti-strange

143 9702/13/O/N/18/Q38

A neutron collides with a nucleus of uranium-235. One possible nuclear reaction that results is represented by the equation



where x represents one or more particles.

What does x represent?

- A one neutron
- B two electrons
- C two neutrons
- D two protons

144 9702/13/O/N/18/Q39

A nucleus Q has the notation ${}_x^y\text{Q}$.

Which of the following is an isotope of Q?

- A ${}_x^{y-1}\text{Q}$ B ${}_{x-1}^y\text{Q}$ C ${}_{x+1}^y\text{Q}$ D ${}_{x+1}^{y-1}\text{Q}$

145 9702/13/O/N/18/Q40

In β^- decay, a neutron inside a nucleus changes to a proton.

Which statement describes the quark composition of the nucleus during the decay?

- A The number of down quarks decreases by one.
- B The number of down quarks increases by one.
- C The number of down quarks stays the same.
- D The number of up quarks stays the same.

146 9702/11/M/J/19/Q39

Which statement about the alpha-particle scattering experiment provides evidence for the existence of the nucleus?

- A A tiny proportion of the alpha-particles are deflected through large angles.
- B Slower alpha-particles are deflected through larger angles.
- C The kinetic energies of the deflected alpha-particles are unchanged.
- D The number of alpha-particles deflected depends on the thickness of the foil.

147 9702/11/M/J/19/Q40

Some particles are a combination of three quarks.

Which combination of quarks would **not** result in a particle with a charge of either $+1.6 \times 10^{-19} \text{ C}$ or zero?

- A** up, down, down
- B** up, strange, strange
- C** up, up, down
- D** up, up, up

148 9702/12/M/J/19/Q38

A nucleus of francium-221 ($^{221}_{87}\text{Fr}$) decays into a nucleus of bismuth-209 ($^{209}_{83}\text{Bi}$) in several steps.

Which particles could be emitted?

- A** 2 α -particles and 4 β^- particles
- B** 2 α -particles and 4 β^+ particles
- C** 3 α -particles and 2 β^- particles
- D** 3 α -particles and 2 β^+ particles

149 9702/12/M/J/19/Q39

Which equation describes the changes to the quark composition of a nucleus and the lepton emission during the process of β^+ decay?

- A** down $\rightarrow$ up + positron + electron neutrino
- B** down $\rightarrow$ up + positron + electron antineutrino
- C** up $\rightarrow$ down + positron + electron neutrino
- D** up $\rightarrow$ down + positron + electron antineutrino

150 9702/12/M/J/19/Q40

There are protons, neutrons and electrons in the simple model of an atom.

To which class (group), hadron or lepton, do these particles belong?

	hadron	lepton
A	electron	proton and neutron
B	neutron	proton and electron
C	proton and electron	neutron
D	proton and neutron	electron

151 9702/13/M/J/19/Q39

What is a correct estimate of the order of magnitude of the diameter of a typical atomic nucleus?

- A** 10^{-14} m **B** 10^{-18} m **C** 10^{-22} m **D** 10^{-26} m

152 9702/13/M/J/19/Q40

Which statement describes β^- decay in terms of a simple quark model?

- A** A down quark changes to an up quark, and an electron and an electron antineutrino are emitted.
B A down quark changes to an up quark, and a positron and an electron neutrino are emitted.
C An up quark changes to a down quark, and an electron and an electron antineutrino are emitted.
D An up quark changes to a down quark, and a positron and an electron neutrino are emitted.

153 9702/11/O/N/19/Q39

An unstable nucleus of an element decays by emitting an α -particle or a β^- particle to become a nucleus of a different element. This nucleus is also unstable and also emits an α -particle or a β^- particle. The process continues until an isotope of the original element is produced.

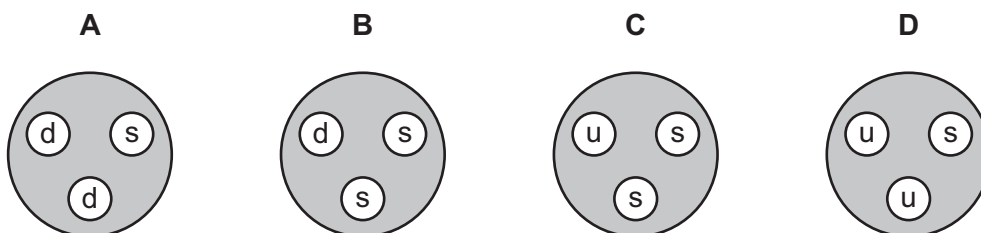
What is the minimum possible number of these particles emitted?

- A** 2 **B** 3 **C** 4 **D** 5

154 9702/11/O/N/19/Q40

The diagrams show the quark composition of four different hadrons. One of the hadrons is a Σ^+ particle. It has a charge of $+e$, where e is the elementary charge.

Which hadron could be the Σ^+ particle?



key
u = up quark
d = down quark
s = strange quark

155 9702/12/O/N/19/Q38

The table gives some data relating to four neutral (uncharged) atoms W, X, Y and Z.

	W	X	Y	Z
nucleon (mass) number	16	17	17	18
total number of particles (protons, neutrons and electrons) in the atom	24	26	25	28

Two of the atoms are isotopes of the same element.

What is the proton number of this element?

- A** 7 **B** 8 **C** 9 **D** 10

156 9702/12/O/N/19/Q39

What is **not** a fundamental particle?

- A** electron
B neutrino
C neutron
D positron

157 9702/12/O/N/19/Q40

An unstable nucleus decays and emits a β^- particle.

Which changes, if any, occur to the quark composition of the nucleus?

	quark changes	
	up quarks	down quarks
A	+1	0
B	+1	-1
C	-1	+1
D	0	+1

158 9702/13/O/N/19/Q39

A radioactive nucleus emits an α -particle or a β^- particle, creating a product nucleus.

Which decay could create the product nucleus stated?

	radioactive nucleus	decay	product nucleus
A	$^{226}_{88}\text{Ra}$	α	$^{224}_{86}\text{Rn}$
B	$^{238}_{92}\text{U}$	α	$^{242}_{94}\text{Pu}$
C	$^{228}_{88}\text{Ra}$	β^-	$^{228}_{87}\text{Fr}$
D	$^{231}_{90}\text{Th}$	β^-	$^{231}_{91}\text{Pa}$

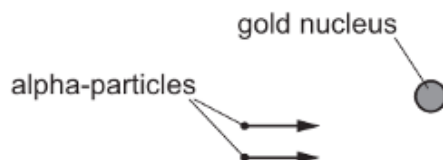
159 9702/13/O/N/19/Q40

Which statement is correct?

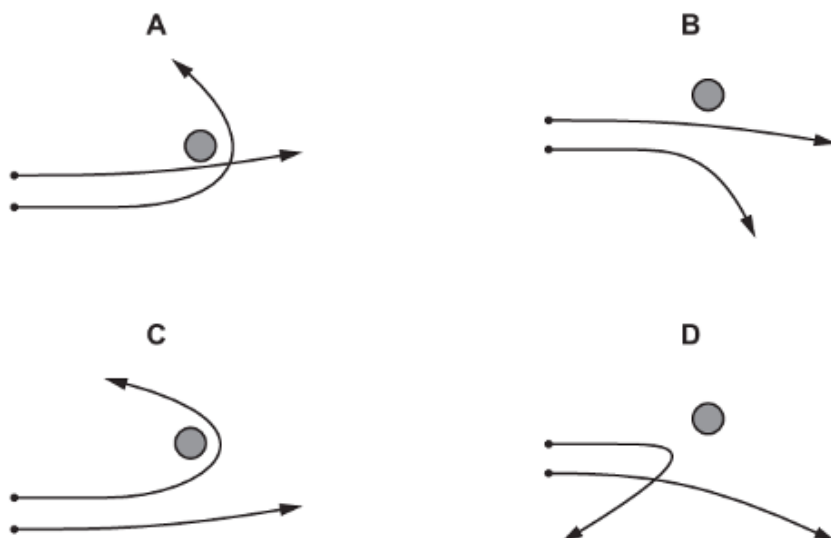
- A** Electrons and neutrinos are fundamental particles.
- B** Electrons and neutrinos are hadrons.
- C** Protons and neutrons are leptons.
- D** Protons and neutrons are quarks.

160. 9702/12/F/M/20/Q38

Two alpha-particles with the same kinetic energy are moving towards, and are then deflected by, a gold nucleus.

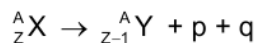


Which diagram could show the paths of the two alpha-particles?



161. 9702/12/F/M/20/Q39

The equation represents the decay of a nucleus X to a nucleus Y.



What are particles p and q?

	p	q
A	β^- particle	neutron
B	β^- particle	proton
C	β^+ particle	antineutrino
D	β^+ particle	neutrino

162. 9702/12/F/M/20/Q40

Which row gives the correct type and quark composition for the named particle?

	particle	type	quark composition
A	neutron	hadron	u u d
B	neutron	lepton	u d d
C	proton	hadron	u u d
D	proton	lepton	u d d

key

u = up quark

d = down quark

163. 9702/11/M/J/20/Q39

An element has two isotopic forms.

What are the nuclear arrangements of these two isotopes?

- A** They have different nucleon numbers and different proton numbers.
- B** They have different nucleon numbers but the same proton number.
- C** They have the same nucleon number and the same proton number.
- D** They have the same nucleon number but different proton numbers.

164. 9702/11/M/J/20/Q40

A hadron has a charge $+e$, where e is the elementary charge.

Which combination of up (u) and down (d) quarks could form this hadron?

- A** ddd
- B** udd
- C** uud
- D** uuu

165. 9702/12/M/J/20/Q39

An unstable nucleus goes through successive decays to become a final, stable nucleus.

The initial nucleus and the final nucleus are isotopes of each other.

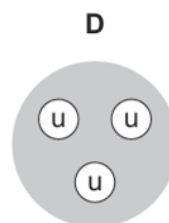
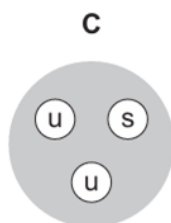
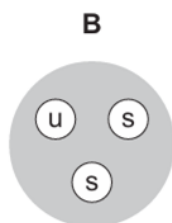
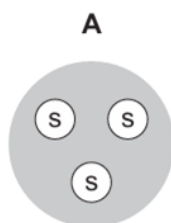
How many α and β^- particles could have been emitted during the decay sequence?

	particle	
	α	β^-
A	1	0
B	1	2
C	2	0
D	2	1

166. 9702/12/M/J/20/Q40

A hadron has a charge of $-e$ and is composed of three quarks.

What could be the quark composition of the hadron?



key

u = up quark
s = strange quark

167. 9702/13/M/J/20/Q39

Radiation from a radioactive source has a range of a few millimetres in air and can be deflected by an electric field.

Which type of radiation is being emitted?

- A** α -radiation
- B** β^- radiation
- C** β^+ radiation
- D** γ -rays

168. 9702/13/M/J/20/Q40

Which equation describes the process of β^+ decay?

key:

u = up quark

d = down quark

ν = (electron) neutrino

$\bar{\nu}$ = (electron) antineutrino

A $ddu \rightarrow uud + \beta^+ + \nu$

B $ddu \rightarrow uud + \beta^+ + \bar{\nu}$

C $uud \rightarrow ddu + \beta^+ + \nu$

D $uud \rightarrow ddu + \beta^+ + \bar{\nu}$

169. 9702/11/O/N/20/Q38

Which statement about two nuclei that are isotopes of the same element is correct?

A The nuclei each have the same acceleration when in the same uniform electric field.

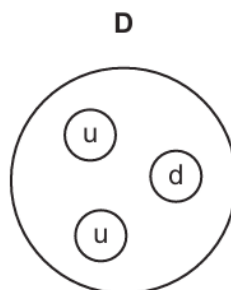
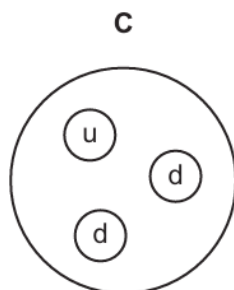
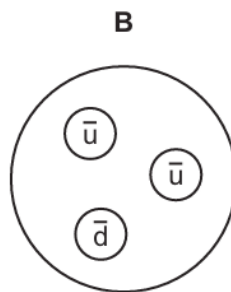
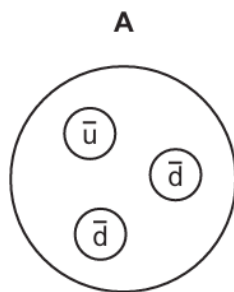
B The nuclei each have the same number of neutrons.

C The nuclei each have the same number of nucleons.

D Uncharged atoms containing the nuclei each have the same number of electrons.

170. 9702/11/O/N/20/Q40

Which diagram represents the quark composition of an antineutron?



key

u up quark

d down quark

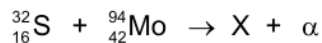
$\bar{u}$ up antiquark

$\bar{d}$ down antiquark

171. 9702/11/O/N/20/Q39

In a nuclear physics experiment, a nucleus of $^{32}_{16}\text{S}$ collides with a nucleus of $^{94}_{42}\text{Mo}$. The nuclei combine together and immediately emit a single alpha-particle.

The nuclear reaction is shown.



What is nucleus X?

A $^{122}_{56}\text{X}$

B $^{124}_{54}\text{X}$

C $^{126}_{58}\text{X}$

D $^{126}_{59}\text{X}$

172. 9702/12/O/N/20/Q39

What is a conclusion from the alpha-particle scattering experiment?

A Protons and electrons have equal but opposite charges.

B Protons have a much larger mass than electrons.

C The nucleus contains most of the mass of the atom.

D The nucleus of an atom contains protons and neutrons.

173. 9702/12/O/N/20/Q40

Which particle is a hadron?

A electron

B neutrino

C positron

D proton

174. 9702/13/O/N/20/Q39

The table contains data for four different nuclei P, Q, R and S.

nucleus	number of neutrons	nucleon number
P	5	10
Q	6	10
R	6	14
S	8	16

Which two nuclei are isotopes of the same element?

A P and Q

B P and S

C Q and R

D R and S

175. 9702/13/O/N/20/Q40

Which particle is a fundamental particle?

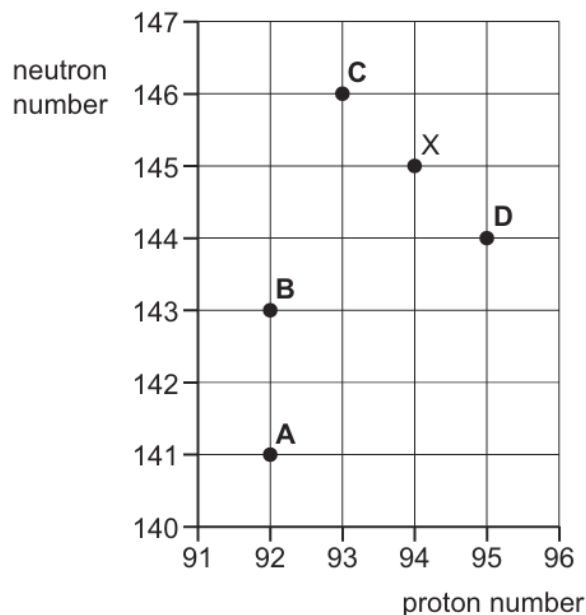
- A alpha-particle
- B electron
- C neutron
- D proton

176. 9702/12/F/M/21/Q39

The figure shows part of a chart of nuclides where neutron number is plotted against proton number.

An unstable nuclide X decays by emitting an α -particle.

Which nuclide is formed by the decay of nuclide X?

**177. 9702/12/F/M/21/Q40**

The nuclei of common isotopes of hydrogen, helium, lithium and beryllium are shown.

Which nucleus contains equal numbers of up and down quarks?

- A ${}^1_1\text{H}$
- B ${}^4_2\text{He}$
- C ${}^7_3\text{Li}$
- D ${}^9_4\text{Be}$

178. 9702/11/M/J/21/Q39

A nucleus of uranium, ${}^{235}_{92}\text{U}$, undergoes a series of decays. During the series of decays, two α -particles and one β^- particle are emitted.

As a result, a nucleus of actinium, Ac, is formed.

What is the correct notation for the nuclide of actinium that is formed?

- A ${}^{227}_{87}\text{Ac}$
- B ${}^{227}_{89}\text{Ac}$
- C ${}^{231}_{87}\text{Ac}$
- D ${}^{231}_{89}\text{Ac}$

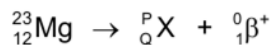
179. 9702/11/M/J/21/Q40

Which particle is a fundamental particle?

- A electron
- B hadron
- C neutron
- D proton

180. 9702/12/M/J/21/Q39

A nucleus of magnesium decays into a nucleus X by emitting a β^+ particle. The decay is represented by the equation shown.



What are the values of P and Q?

	P	Q
A	22	11
B	22	13
C	23	11
D	23	13

181. 9702/12/M/J/21/Q40

In β^- decay, a neutron inside a nucleus changes to a proton.

Which statement describes the quark composition of the nucleus during the decay?

- A The number of down quarks decreases by one.
- B The number of down quarks increases by one.
- C The number of down quarks stays the same.
- D The number of up quarks stays the same.

182. 9702/13/M/J/21/Q39

A beam of α -particles is incident on a thin gold foil. One α -particle collides head-on with a gold nucleus and is deflected back along its original path.

Which statement could explain why the recoil speed of the gold nucleus is small compared with the recoil speed of the α -particle?

- A Most α -particles are only slightly deflected as they pass through the gold foil.
- B The α -particle and the gold nucleus repel each other.
- C The mass of the gold nucleus is much greater than the mass of the α -particle.
- D The momentum of the α -particle decreases as it approaches the gold nucleus.

183. 9702/13/M/J/21/Q40

A hadron is composed of three quarks. The hadron has a charge.

What is a possible quark composition of the hadron?

- A down, down, up
- B down, up, strange
- C up, strange, strange
- D up, up, strange

184. 9702/11/O/N/21/Q39

A uranium-238 nucleus, $^{238}_{92}\text{U}$, undergoes a series of nuclear decays to form uranium-234, $^{234}_{92}\text{U}$.

Which series of decays could give this result?

- A emission of four β^- particles
- B emission of four γ -rays
- C emission of one α -particle and two β^- particles
- D emission of two α -particles and eight β^- particles

185. 9702/11/O/N/21/Q40

Which combination of up (u) and down (d) quarks forms a proton?

- A u u u B u u d C u d d D d d d

186. 9702/12/O/N/21/Q39

An unstable nucleus decays by emitting a β^+ particle.

Which statement is correct?

- A An antineutrino is also emitted.
- B A neutron changes into a proton.
- C Mass–energy is conserved.
- D The nucleon number is not conserved.

187. 9702/12/O/N/21/Q40

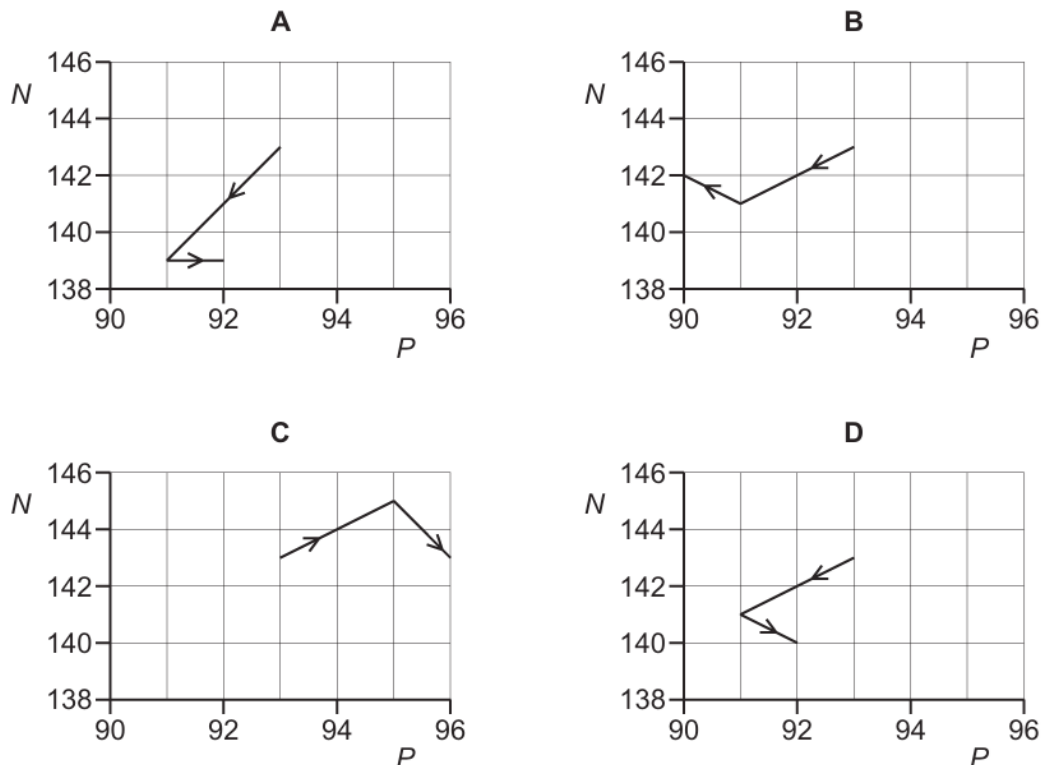
Which statement is **not** correct?

- A An antineutrino is a fundamental particle.
- B An electron is made up of a quark and an antiquark.
- C A neutrino is a lepton.
- D A neutron is composed of three quarks.

188. 9702/13/O/N/21/Q39

A nucleus of neptunium-236 contains 93 protons and 143 neutrons. This nucleus decays with the emission of an α -particle. The nucleus formed then emits a β^- particle.

Which diagram shows the changes in the number P of protons and the number N of neutrons in these nuclei?



189. 9702/13/O/N/21/Q40

Which combination of quarks could **not** be the quark composition of the hadron shown?

p = proton

n = neutron

Σ^- = sigma particle of charge $-e$

Σ^+ = sigma particle of charge $+e$

u = up quark

d = down quark

s = strange quark

	hadron	quark composition
A	Σ^-	dds
B	Σ^+	uds
C	p	uud
D	n	udd

190. 9702/13/O/N/21/Q38

When α -particles are fired at a thin metal foil, most of the particles pass straight through but a few are deflected by a large angle.

Which change would increase the **proportion** of α -particles deflected by a large angle?

- A** using α -particles with greater kinetic energy
- B** using a double thickness foil
- C** using a foil made of a metal with fewer protons in its nuclei
- D** using a source emitting more α -particles per unit time

191. 9702/12/F/M/22/Q38

When α -particles are directed at gold leaf:

- 1 almost all α -particles pass through without deflection
- 2 a few α -particles are deviated through large angles.

What are the reasons for these effects?

	1	2
A	most α -particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α -particles bounce back from the gold surface
B	most α -particles miss all gold atoms	a few α -particles bounce off gold atoms
C	the gold nucleus is very small so most α -particles miss all nuclei	occasionally the path of an α -particle is close to a nucleus
D	the positive charge in an atom is not concentrated enough to deflect an α -particle	occasionally an α -particle experiences many small deflections in the same direction

192. 9702/12/F/M/22/Q39

A nucleus X is radioactive and decays into a nucleus Y.

X and Y are isotopes of the same element.

Which combination of particles could have been emitted during the decay process?

- A** 1 α -particle and 1 β^- particle
- B** 1 α -particle and 2 β^- particles
- C** 2 α -particles and 1 β^- particle
- D** 2 α -particles and 2 β^- particles

193. 9702/12/F/M/22/Q40

A positively charged meson consists of a quark and an antiquark.

What could be the quark and antiquark?

- A charm and antiup
- B down and antitop
- C strange and antibottom
- D up and antistrange

194. 9702/11/M/J/22/Q38

Carbon-14 decays into nitrogen-14 by emitting a β^- particle.

Which statement explains why the β^- particles are emitted with a range of different kinetic energies?

- A The carbon-14 nuclei have slightly different masses.
- B The emitted β^- particles have a range of different masses.
- C The energy released in the decay process is different for each carbon-14 nucleus that decays.
- D The energy released in the decay process is shared between the nitrogen-14 nucleus, a β^- particle and an antineutrino.

195. 9702/11/M/J/22/Q39

A nucleus of a radioactive element emits an α -particle, then a β^- particle and then another β^- particle.

Which statement describes the final element that is produced?

- A It is a different element of higher proton number than the original element.
- B It is a different element of lower nucleon number than the original element.
- C It is an isotope of the original element.
- D It is the same element as the original element but with a different proton number.

196. 9702/11/M/J/22/Q40

How many hadrons, baryons and mesons are there in a nucleus of ${}^9_4\text{Be}$?

	hadrons	baryons	mesons
A	9	4	5
B	9	5	4
C	9	9	0
D	13	9	0

197. 9702/12/M/J/22/Q38

In an experiment on α -particle scattering, α -particles are directed at a thin gold foil. Most of the α -particles pass straight through the foil or are deflected by a small angle. A small number of α -particles are deflected by a large angle.

Which statement **cannot** be deduced from this experiment?

- A Atoms are mostly empty space.
- B Most of the mass of an atom is concentrated in the nucleus.
- C The nucleus of an atom contains protons.
- D The nucleus of an atom is small compared to the size of an atom.

198. 9702/12/M/J/22/Q39

Four nuclei are represented below.



Which statement about these nuclei is correct?

- A An uncharged atom of element Q has 24 orbital electrons.
- B Nuclei G and M are isotopes of the same element.
- C When E absorbs a neutron and then emits an α -particle, E transforms into M.
- D When M emits a β^- particle, M transforms into Q.

199. 9702/12/M/J/22/Q40

A neutron is composed of one up (u) quark and two down (d) quarks. When a neutron decays to a proton, a beta-particle is emitted.

What is the change in the quark structure of the neutron due to the emission of the beta-particle?

(The symbol for a neutrino is ν_e and for an antineutrino is $\bar{\nu}_e$.)

- A $d \rightarrow u + \beta^- + \nu_e$
- B $d \rightarrow u + \beta^- + \bar{\nu}_e$
- C $u \rightarrow d + \beta^+ + \nu_e$
- D $u \rightarrow d + \beta^+ + \bar{\nu}_e$

200. 9702/13/M/J/22/Q40

Which statement is **not** correct?

- A A meson consists of three quarks.
- B A proton is a baryon.
- C A quark is a fundamental particle.
- D There are six flavours (types) of quark.

201. 9702/13/M/J/22/Q39

An actinium nucleus has a nucleon number of 227 and a proton number of 89. It decays to form a radium nucleus, emitting a β^- particle and an α -particle in the process.

What are the nucleon number and the proton number of this radium nucleus?

	nucleon number	proton number
A	223	87
B	223	88
C	224	87
D	225	86

202. 9702/11/O/N/22/Q38

An α -particle passes close to a gold nucleus and is deflected through an angle greater than 90° .

Which property of the α -particle changes as a result of the deflection?

- A** charge
- B** momentum
- C** nucleon number
- D** proton number

203. 9702/11/O/N/22/Q39

A nucleus of ${}_{92}^{238}\text{U}$ decays in stages by emitting α -particles and β^- particles, eventually forming a nucleus of ${}_{82}^{206}\text{Pb}$.

How many α -particles and how many β^- particles are emitted during the decay chain?

	α -particles	β^- particles
A	8	6
B	8	10
C	16	6
D	16	22

204. 9702/11/O/N/22/Q40

Which list of particles contains **only** fundamental particles?

- A** antineutrino, beta, meson
- B** baryon, neutrino, positron
- C** electron, neutrino, alpha
- D** lepton, quark, positron

205. 9702/12/O/N/22/Q38

In the α -particle scattering experiment, a beam of α -particles is aimed at a thin gold foil. Most of the α -particles go straight through or are deflected by a small angle. A very small proportion are deflected by more than 90° , effectively rebounding towards the source of the α -particles.

Which conclusion about the structure of atoms **cannot** be drawn from this experiment alone?

- A Most of the atom is empty space.
- B Most of the mass of an atom is concentrated in the nucleus.
- C The nucleus contains both protons and neutrons.
- D The nucleus is charged.

206. 9702/12/O/N/22/Q39

Which two particles have opposite charges?

- A alpha-particle and helium nucleus
- B antiproton and beta-plus particle
- C beta-minus particle and electron
- D positron and proton

207. 9702/12/O/N/22/Q40

Which particle is a lepton?

- A meson
- B positron
- C proton
- D quark

208. 9702/13/O/N/22/Q38

Which statement about the alpha-particle scattering experiment provides evidence for the existence of the nucleus?

- A A tiny proportion of the alpha-particles are deflected through large angles.
- B Slower alpha-particles are deflected through larger angles.
- C The kinetic energies of the deflected alpha-particles are unchanged.
- D The number of alpha-particles deflected depends on the thickness of the foil.

209. 9702/13/O/N/22/Q40

A particle consists of two up quarks and a charm quark.

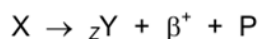
What is the charge of this particle, expressed in terms of the elementary charge e ?

- A $-2e$
- B $-e$
- C $+e$
- D $+2e$

210. 9702/13/O/N/22/O39

A nucleus X undergoes β^+ decay.

The products are a nucleus Y with proton number Z, a β^+ particle and another particle P.



What is particle P and what is the proton number of nucleus X?

	P	proton number of nucleus X
A	antineutrino	$Z - 1$
B	antineutrino	$Z + 1$
C	neutrino	$Z - 1$
D	neutrino	$Z + 1$

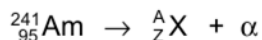
211. 9702/12/F/M/23/Q38

What is a conclusion from the alpha-particle scattering experiment?

- A** Protons and electrons have equal but opposite charges.
- B** Protons have a much larger mass than electrons.
- C** The nucleus contains most of the mass of the atom.
- D** The nucleus of an atom contains protons and neutrons.

212. 9702/12/F/M/23/Q39

Americium-241 is a radioactive nuclide used in smoke detectors. It undergoes α -decay to form nuclide X. This decay may be represented by the equation shown.



What are the values of A and Z?

	A	Z
A	237	93
B	239	91
C	241	94
D	241	96

213. 9702/12/F/M/23/Q40

A top quark has a charge of $+\frac{2}{3}e$, where e is the elementary charge.

What is the charge of an anti top quark?

- A** $-\frac{2}{3}e$
- B** $-\frac{1}{3}e$
- C** $+\frac{1}{3}e$
- D** $+\frac{2}{3}e$

214. 9702/11/M/J/23/Q38

A proton has mass m_p and charge $+e$.

What are the mass and charge of an antiproton?

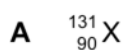
	mass	charge
A	$-m_p$	$+e$
B	$-m_p$	$-e$
C	m_p	$+e$
D	m_p	$-e$

215. 9702/11/M/J/23/Q39

A uranium nucleus has 92 protons and 143 neutrons.

The nucleus emits a total of 3 α -particles and 4 β^- particles to form nucleus X.

How can nucleus X be represented?

**216. 9702/11/M/J/23/Q40**

Which statement about a proton is correct?

- A** A proton is a baryon but not a meson.
- B** A proton is a meson but not a hadron.
- C** A proton is both a hadron and a meson.
- D** A proton is both a meson and a baryon.

217. 9702/12/M/J/23/Q38

What are isotopes?

- A** nuclei of different elements with the same number of neutrons
- B** nuclei of different elements with the same number of nucleons
- C** nuclei of the same element with different numbers of neutrons
- D** nuclei of the same element with different numbers of protons

218. 9702/12/M/J/23/Q39

A neutron ${}_0^1\text{n}$ is fired at a ${}_{92}^{235}\text{U}$ nucleus. The neutron is absorbed by the nucleus which then splits to form nuclei of ${}_{56}^{141}\text{Ba}$ and ${}_{36}^{92}\text{Kr}$.

What is the number of free neutrons emitted when the ${}_{92}^{235}\text{U}$ nucleus splits?

- A** 0
- B** 1
- C** 2
- D** 3

219. 9702/12/M/J/23/Q40

A π^+ meson has a charge of $+e$, where e is the elementary charge. It consists of an up quark and one other quark.

What could be the other quark in the π^+ meson?

- A** anti-down
- B** anti-up
- C** bottom
- D** charm

220. 9702/13/M/J/23/Q38

The table contains data for four different nuclei P, Q, R and S.

nucleus	number of neutrons	nucleon number
P	5	10
Q	6	10
R	6	14
S	8	16

Which two nuclei are isotopes of the same element?

- A** P and Q
- B** P and S
- C** Q and R
- D** R and S

221. 9702/13/M/J/23/Q39

Which statement about β^- decay is correct?

- A** A neutron changes to a proton in the nucleus and an electron and an antineutrino are emitted.
- B** A neutron changes to a proton in the nucleus and an electron and a neutrino are emitted.
- C** A proton changes to a neutron in the nucleus and an electron and an antineutrino are emitted.
- D** A proton changes to a neutron in the nucleus and an electron and a neutrino are emitted.

222. 9702/13/M/J/23/Q40

What is the quark composition of a hydrogen-3 nucleus, ${}^3_1\text{H}$?

	number of quarks	
	up	down
A	4	5
B	5	4
C	5	7
D	7	5

223. 9702/11/O/N/23/Q39

Which nuclide is formed when $^{10}_6\text{C}$ undergoes β^+ decay?

- A** $^{11}_6\text{C}$ **B** ^9_6C **C** $^{10}_5\text{B}$ **D** $^{10}_7\text{N}$

224. 9702/11/O/N/23/Q40

A particular hadron is composed of three quarks and has zero charge.

What is a possible quark composition of the hadron?

- A** down, down, strange
B up, down, strange
C up, up, down
D up, up, strange

225. 9702/12/O/N/23/Q38

Two nuclides are different isotopes of the same element.

Which statement about the nuclides is correct?

- A** Neutral atoms of the nuclides have the same number of electrons.
- B** Nuclei of the nuclides have different numbers of protons.
- C** Nuclei of the nuclides have the same number of nucleons.
- D** Nuclei of the nuclides have the same number of neutrons.

226. 9702/12/O/N/23/Q39

The charge-to-mass ratio r of a particle is given by the equation shown.

$$r = \frac{\text{charge on particle}}{\text{mass of particle}}$$

The value of r is determined for an α -particle, a β^+ particle and a proton p.

Which list shows the particles in order of increasing magnitude of r from left to right?

- A** $\alpha \rightarrow \beta^+ \rightarrow p$
B $\alpha \rightarrow p \rightarrow \beta^+$
C $p \rightarrow \alpha \rightarrow \beta^+$
D $p \rightarrow \beta^+ \rightarrow \alpha$

227. 9702/12/O/N/23/Q40

Which combination of up (u) and down (d) quarks forms a neutron?

- A** uuu **B** uud **C** udd **D** ddd

228. 9702/13/O/N/23/Q38

The table shows the number of nucleons and the total number of particles (protons, neutrons and electrons) in neutral atoms of four nuclides W, X, Y and Z.

	number of nucleons	total number of particles
W	19	30
X	19	31
Y	21	32
Z	22	31

Which two nuclides are isotopes of each other?

- A** W and X **B** W and Y **C** X and Z **D** Y and Z

229. 9702/13/O/N/23/Q39

When a sample of a radioactive isotope decays by α -particle emission, the α -particles emitted have a single discrete energy.

When a sample of a radioactive isotope decays by β^- particle emission, the β^- particles emitted have a continuous range of energies.

What is the explanation for this?

- A** An antineutrino is emitted with a β^- particle but not with an α -particle.
B An antineutrino is emitted with an α -particle but not with a β^- particle.
C The α -particles have much more energy than the β^- particles.
D The β^- particles have much more energy than the α -particles.

230. 9702/13/O/N/23/Q40

Some particles are a combination of three quarks.

Which combination of quarks does **not** result in a particle with a charge of either $+1.6 \times 10^{-19} \text{ C}$ or zero?

- A** up, down, down
B up, strange, strange
C up, up, down
D up, up, up

ANSWERS

1. C	46. B	91. C	136. B	179. A	222. A
2. A	47. B	92. A	137. D	180. C	223. C
3. C	48. C	93. D	138. B	181. A	224. B
4. A	49. C	94. A	139. D	182. C	225. A
5. A	50. D	95. C	140. B	183. D	226. B
6. C	51. C	96. B	141. C	184. C	227. C
7. C	52. D	97. B	142. A	185. B	228. B
8. B	53. C	98. A	143. C	186. C	229. A
9. D	54. B	99. C	144. A	187. B	230. D
10. C	55. A	100. C	145. A	188. D	
11. B	56. C	101. C	146. A	189. B	
12. C	57. C	102. C	147. D	190. B	
13. C	58. C	103. D	148. C	191. C	
14. D	59. A	104. D	149. C	192. B	
15. D	60. C	105. B	150. D	193. D	
16. C	61. D	106. C	151. A	194. D	
17. C	62. C	107. A	152. A	195. C	
18. D	63. B	108. C	153. B	196. C	
19. C	64. D	109. C	154. D	197. C	
20. B	65. C	110. C	155. B	198. C	
21. C	66. B	111. B	156. C	199. B	
22. D	67. D	112. A	157. B	200. A	
23. B	68. B	113. C	158. D	201. B	
24. D	69. D	114. C	159. A	202. B	
25. A	70. C	115. D	160. D	203. A	
26. D	71. B	116. A	161. D	204. D	
27. B	72. D	117. B	162. C	205. C	
28. C	73. C	118. A	163. B	206. B	
29. D	74. C	119. A	164. C	207. B	
30. B	75. B	120. A	165. B	208. A	
31. A	76. B	121. B	166. A	209. D	
32. D	77. B	122. C	167. A	210. D	
33. C	78. A	123. B	168. C	211. C	
34. C	79. C	124. A	169. D	212. A	
35. C	80. D	125. A	170. A	213. A	
36. C	81. A	126. C	171. A	214. D	
37. B	82. C	127. C	172. C	215. D	
38. A	83. C	128. A	173. D	216. A	
39. D	84. A	129. C	174. D	217. C	
40. D	85. C	130. D	175. B	218. D	
41. A	86. B	131. D	176. B	219. A	
42. A	87. B	132. C	177. B	220. D	
43. C	88. B	133. B	178. B	221. A	
44. B	89. D	134. B			
45. A	90. B	135. C			

- 1 The spontaneous and random decay of a radioactive substance involves the emission of either α -radiation or β -radiation and/or γ -radiation.

(a) Explain what is meant by *spontaneous* decay.

.....
.....
..... [2]

(b) State the type of emission, one in each case, that

(i) is not affected by electric and magnetic fields,

..... [1]

(ii) produces the greatest density of ionisation in a medium,

..... [1]

(iii) does not directly result in a change in the proton number of the nucleus,

..... [1]

(iv) has a range of energies, rather than discrete values.

..... [1]

- 2 An α -particle A approaches and passes by a stationary gold nucleus N. The path is illustrated in Fig. 7.1.

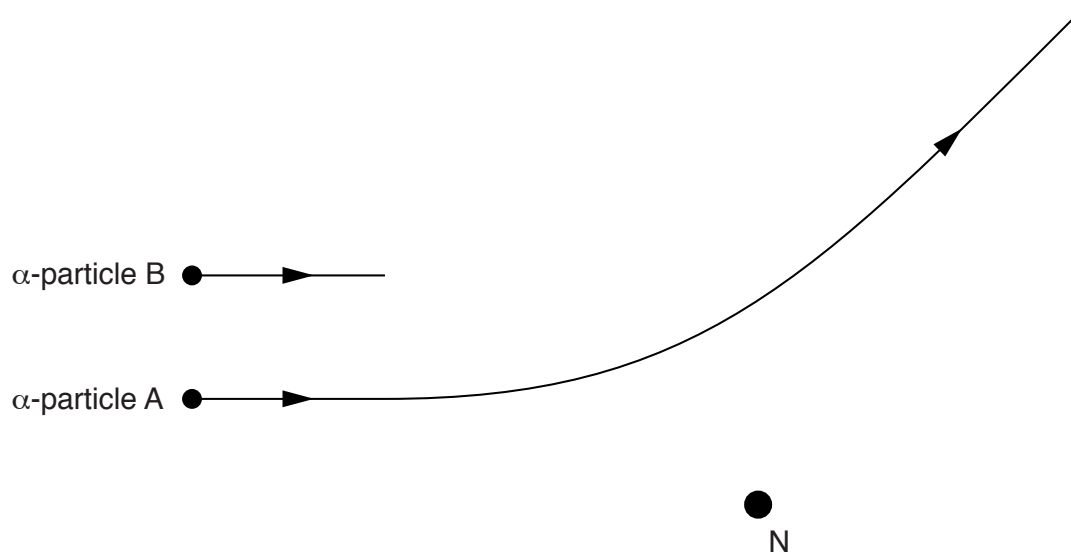


Fig. 7.1

- (a) On Fig. 7.1, mark the angle of deviation D of this α -particle as a result of passing the nucleus N. [1]
- (b) A second α -particle B has the same initial direction and energy as α -particle A. On Fig. 7.1, complete the path of α -particle B as it approaches and passes by the nucleus N. [2]
- (c) State what can be inferred about atoms from the observation that very few α -particles experience large deviations.

.....

.....

..... [2]

- (d) The nucleus N could be one of several different isotopes of gold.

Suggest, with an explanation, whether different isotopes of gold would give rise to different deviations of a particular α -particle.

.....

.....

..... [2]

- 3 Tungsten-184 ($^{184}_{74}\text{W}$) and tungsten-185 ($^{185}_{74}\text{W}$) are two isotopes of tungsten.

Tungsten-184 is stable but tungsten-185 undergoes β -decay to form rhenium (Re).

- (a) Explain what is meant by *isotopes*.

.....

 [2]

- (b) The β -decay of nuclei of tungsten-185 is spontaneous and random.

State what is meant by

- (i) *spontaneous* decay,

.....
 [1]

- (ii) *random* decay.

.....
 [1]

- (c) Complete the nuclear equation for the β -decay of a tungsten-185 nucleus.



4 One of the isotopes of uranium is uranium-238 (${}^{238}_{92}\text{U}$).

9702/21/M/J/10/Q7

(a) State what is meant by *isotopes*.

.....
..... [2]

(b) For a nucleus of uranium-238, state

(i) the number of protons,

number = [1]

(ii) the number of neutrons.

number = [1]

(c) A uranium-238 nucleus has a radius of $8.9 \times 10^{-15} \text{ m}$.

Calculate, for a uranium-238 nucleus,

(i) its mass,

mass = kg [2]

(ii) its mean density.

density = kg m^{-3} [2]

(d) The density of a lump of uranium is $1.9 \times 10^4 \text{ kg m}^{-3}$.

Using your answer to (c)(ii), suggest what can be inferred about the structure of the atom.

.....
.....
..... [2]

- 5 (a) The radioactive decay of some nuclei gives rise to the emission of α -particles.
State

(i) what is meant by an α -particle,

.....[1]

(ii) two properties of α -particles.

1.

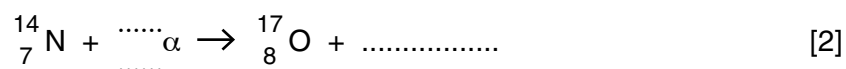
.....

2.

.....[2]

- (b) One possible nuclear reaction involves the bombardment of a stationary nitrogen-14 nucleus by an α -particle to form oxygen-17 and another particle.

(i) Complete the nuclear equation for this reaction.



- (ii) The total mass-energy of the nitrogen-14 nucleus and the α -particle is less than that of the particles resulting from the reaction. This mass-energy difference is 1.1 MeV.

1. Suggest how it is possible for mass-energy to be conserved in this reaction.

.....

.....[1]

2. Calculate the speed of an α -particle having kinetic energy of 1.1 MeV.

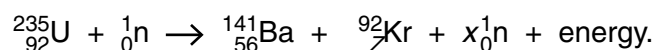
speed = m s⁻¹ [4]

- 6 (a) Uranium (U) has at least fourteen isotopes.
Explain what is meant by *isotopes*.

.....

 [2]

- (b) One possible nuclear reaction involving uranium is



- (i) State three quantities that are conserved in a nuclear reaction.

1.

 2.

 3.
 [3]

- (ii) For this reaction, determine the value of

1. Z ,

$Z =$ [1]

2. x .

$x =$ [1]

- 7 The results of the α -particle scattering experiment provided evidence for the existence and small size of the nucleus.

(a) State the result that provided evidence for

- (i) the small size of the nucleus, compared with the atom,

.....
.....
.....[2]

- (ii) the nucleus being charged and containing the majority of the mass of the atom.

.....
.....
.....[2]

(b) The α -particles in this experiment originated from the decay of a radioactive nuclide.

Suggest two reasons why β -particles from a radioactive source would be inappropriate for this type of scattering experiment.

1.
.....
2.
.....
.....[2]

Do not allow β -particles have negative charge or β -particles have high speed

- 8 (a) Two isotopes of the element uranium are $^{235}_{92}\text{U}$ and $^{238}_{92}\text{U}$.

Explain the term *isotope*.

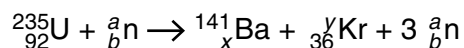
.....

 [2]

- (b) (i) In a nuclear reaction, proton number and neutron number are conserved. Other than proton number and neutron number, state a quantity that is conserved in a nuclear reaction.

..... [1]

- (ii) When a nucleus of uranium-235 absorbs a neutron, the following reaction may take place.



State the values of a , b , x and y .

$a =$

$b =$

$x =$

$y =$

[3]

- (c) When the nucleus of $^{238}_{92}\text{U}$ absorbs a neutron, the nucleus decays, emitting an α -particle. State the proton number and nucleon number of the nucleus that is formed as a result of the emission of the α -particle.

proton number =

nucleon number =

[2]

- 9 (a) State the experimental observations that show radioactive decay is

- (i) spontaneous,

..... [1]

- (ii) random.

..... [1]

- (b) On Fig. 7.1, complete the charge and mass of α -particles, β -particles and γ -radiation. Give example speeds of α -particles and γ -radiation emitted by a laboratory source.

	α -particle	β -particle	γ -radiation
charge			0
mass	4u		
speed		up to $0.99c$	

Fig. 7.1

[3]

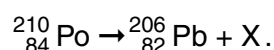
- (c) Explain the process by which α -particles lose energy when they pass through air.

.....

 [2]

9702/21/M/J/12/Q7

- 10 (a) The spontaneous decay of polonium is shown by the nuclear equation



- (i) State the composition of the nucleus of X.

.....
 [1]

- (ii) The nuclei X are emitted as radiation. State two properties of this radiation.

1.

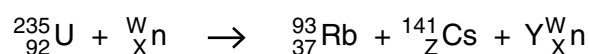
 2.
 [2]

- (b) The mass of the polonium (Po) nucleus is greater than the combined mass of the nuclei of lead (Pb) and X. Use a conservation law to explain qualitatively how this decay is possible.

.....

 [3]

- 11 (a) A nuclear reaction occurs when a uranium-235 nucleus absorbs a neutron. The reaction may be represented by the equation:



State the number represented by the letter

W Y

X Z [3]

- (b) The sum of the masses on the left-hand side of the equation in (a) is not the same as the sum of the masses on the right-hand side.

Explain why mass seems not to be conserved.

.....
 [2]

- 12 A radioactive source emits α -radiation and γ -radiation.

Explain how it may be shown that the source does not emit β -radiation using

- (a) the absorption properties of the radiation,

.....

 [2]

- (b) the effects of a magnetic field on the radiation.

.....

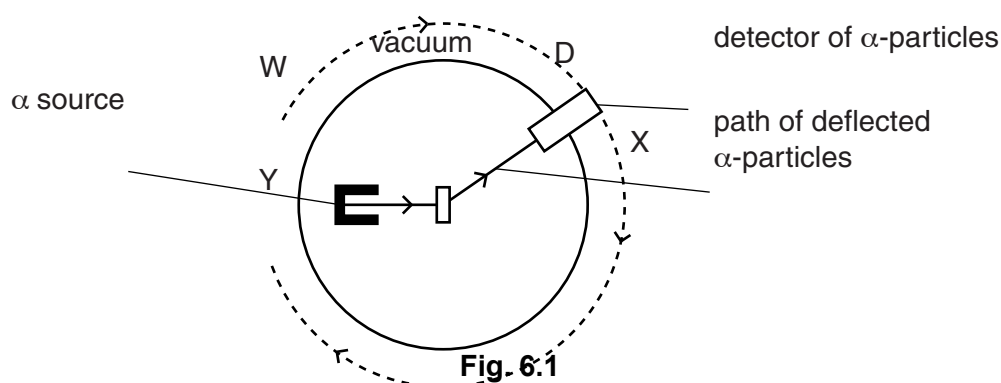
 [2]

13 (a) Describe the structure of an atom of the nuclide $^{235}_{92}\text{U}$.

9702/21/O/N/12/Q6

.....
 [2]

(b) The deflection of α -particles by a thin metal foil is investigated with the arrangement shown in Fig. 6.1. All the apparatus is enclosed in a vacuum.



The detector of α -particles, D, is moved around the path labelled WXY.

(i) Explain why the apparatus is enclosed in a vacuum.

.....
 [1]

(ii) State and explain the readings detected by D when it is moved along WXY.

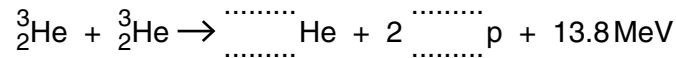
.....

 [3]

(c) A beam of α -particles produces a current of 1.5 pA. Calculate the number of α -particles per second passing a point in the beam.

number = s^{-1} [3]

- 14** A nuclear reaction between two helium nuclei produces a second isotope of helium, two protons and 13.8 MeV of energy. The reaction is represented by the following equation.



- (a)** Complete the nuclear equation. [2]

- (b)** By reference to this reaction, explain the meaning of the term *isotope*.

.....
 [2]

- (c)** State the quantities that are conserved in this nuclear reaction.

.....

 [2]

- (d)** Radiation is produced in this nuclear reaction.

State

- (i)** a possible type of radiation that may be produced,

..... [1]

- (ii)** why the energy of this radiation is less than the 13.8 MeV given in the equation.

..... [1]

- (e)** Calculate the minimum number of these reactions needed per second to produce power of 60 W.

number = s⁻¹ [2]

15 (a) β -radiation is emitted during the spontaneous radioactive decay of an unstable nucleus.

(i) State the nature of a β -particle.

..... [1]

(ii) State two properties of β -radiation.

1.

2. [2]

(iii) Explain the meaning of *spontaneous radioactive decay*.

.....
 [1]

(b) The following equation represents the decay of a nucleus of hydrogen-3 by the emission of a β -particle.

Complete the equation.



(c) The β -particle is emitted with an energy of $5.7 \times 10^3 \text{ eV}$.

Calculate the speed of the β -particle.

speed = ms^{-1} [3]

(d) A different isotope of hydrogen is hydrogen-2 (deuterium). Describe the similarities and differences between the atoms of hydrogen-2 and hydrogen-3.

.....

 [2]

16 (a) Describe the two main results of the α -particle scattering experiment. 9702/21/M/J/13/Q7

result 1:

.....

result 2:

.....

[3]

(b) Relate each of the results in **(a)** with the conclusions that were made about the nature of atoms.

result 1:

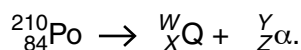
.....

result 2:

.....

[3]

17 A polonium nucleus $^{210}_{84}\text{Po}$ is radioactive and decays with the emission of an α -particle. The nuclear reaction for this decay is given by 9702/22/M/J/13/Q7



(a) (i) State the values of W

X

Y

Z

[2]

(ii) Explain why mass seems not to be conserved in the reaction.

.....

..... [2]

(b) The reaction is spontaneous. Explain the meaning of *spontaneous*.

.....

..... [1]

- 18 (a) An electric field is set up between two parallel metal plates in a vacuum. The deflection of α -particles as they pass between the plates is shown in Fig. 7.1.

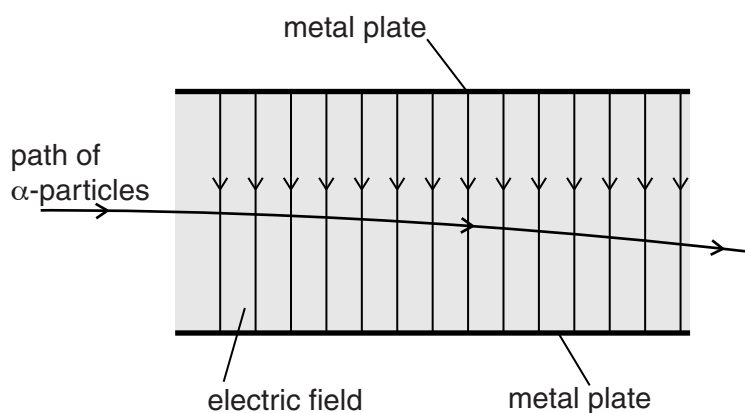


Fig. 7.1

The electric field strength between the plates is reduced. The α -particles are replaced by β -particles. The deflection of β -particles is shown in Fig. 7.2.

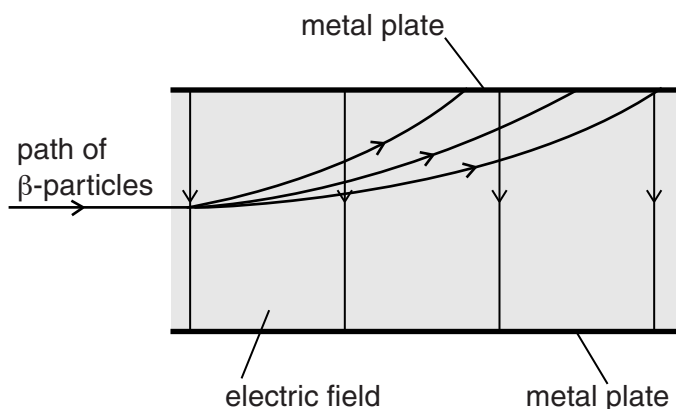


Fig. 7.2

- (i) State one similarity of the electric fields shown in Fig. 7.1 and Fig. 7.2.

.....
[1]

- (ii) The electric field strength in Fig. 7.2 is less than that in Fig. 7.1. State two methods of reducing this electric field strength.

1.
 2.

[2]

- (iii) By reference to the properties of α -particles and β -particles, suggest three reasons for the differences in the deflections shown in Fig. 7.1 and Fig. 7.2.

1.
-
2.
-
3.
-

[3]

- (b) A source of α -particles is uranium-238. The nuclear reaction for the emission of α -particles is represented by



State the values of W

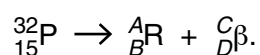
X

Y

Z

[2]

- (c) A source of β -particles is phosphorus-32. The nuclear reaction for the emission of β -particles is represented by



State the values of A

B

C

D

[1]

19 (a) State what is meant by

α -particle:

β -particle:

γ -radiation: [2]

(b) Describe the changes to the proton number and the nucleon number of a nucleus when emission occurs of

(i) an α -particle,

..... [1]

(ii) a β -particle,

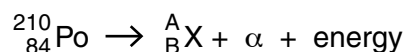
..... [1]

(iii) γ -radiation.

..... [1]

20 In the decay of a nucleus of $^{210}_{84}\text{Po}$, an α -particle is emitted with energy 5.3 MeV.

The emission is represented by the nuclear equation



(a) (i) On Fig. 7.1, complete the number and name of the particle, or particles, represented by A and B in the nuclear equation.

	number	name of particle or particles
A		
B		

Fig. 7.1

[1]

(ii) State the form of energy given to the α -particle in the decay of $^{210}_{84}\text{Po}$.

..... [1]

- (b) A sample of polonium $^{210}_{84}\text{Po}$ emits 7.1×10^{18} α -particles in one day.
 Calculate the mean power output from the energy of the α -particles.

power = W [2]

- 9702/22/M/J/15/Q7

21 A uranium-235 nucleus absorbs a neutron and then splits into two nuclei. A possible nuclear reaction is given by



- (a) State the constituent particles of the uranium-235 nucleus.
 [1]
- (b) Complete Fig. 7.1 for this reaction.

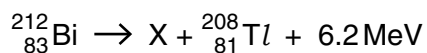
	value
a	
b	
c	
d	

Fig. 7.1

- (c) Suggest a possible form of energy released in this reaction.
 [1]
- (d) Explain, using the law of mass-energy conservation, how energy is released in this reaction.

 [2]

22 The equation represents the spontaneous radioactive decay of a nucleus of bismuth-212.



- (a) (i) Explain the meaning of *spontaneous* radioactive decay.

.....
[1]

- (ii) State the constituent particles of X.

.....[1]

- (b) (i) Use the conservation of mass-energy to explain the release of 6.2MeV of energy in this reaction.

.....

[2]

- (ii) Calculate the energy, in joules, released in this reaction.

energy = J [1]

23 A neutron decays by emitting a β^- particle.

9702/22/F/M/16/Q6

(a) Complete the equation below for this decay.

$${}^1_0\text{n} \rightarrow \begin{array}{c} \text{.....} \\ \text{.....} \end{array} + \begin{array}{c} \text{.....} \\ \text{.....} \end{array} \beta^- + \begin{array}{c} \text{.....} \\ \text{.....} \end{array} \bar{\nu} \quad [2]$$

(b) State the name of the particle represented by the symbol $\bar{\nu}$.

..... [1]

(c) State the name of the class (group) of particles that includes β^- and $\bar{\nu}$.

..... [1]

(d) State

(i) the quark structure of the neutron,

..... [1]

(ii) the change to the quark structure when the neutron decays.

.....

..... [1]

24 (a) Give one example of

a hadron:

a lepton: [1]

(b) Describe, in terms of the simple quark model,

(i) a proton,

.....[1]

(ii) a neutron.

.....[1]

(c) Beta particles may be emitted during the decay of an unstable nucleus of an atom. The emission of a beta particle is due to the decay of a neutron.

(i) Complete the following word equation for the particles produced in this reaction.

neutron $\rightarrow$ + + [1]

(ii) State the change in quark composition of the particles during this reaction.

.....[1]

25 (a) State the name of the class (group) to which each of the following belongs:

electron neutrino

neutron proton [2]

(b) A proton may decay into a neutron together with two other particles.

(i) Complete the following to give an equation that represents this proton decay.

${}^1_1\text{p} \rightarrow \dots\dots \text{n} + \dots\dots + \dots\dots$ [2]

(ii) Write an equation for this decay in terms of quark composition.

(iii) State the name of the force responsible for this decay. [1]

.....[1]

26 (a) Distinguish between an α -particle and a β^+ -particle.

9702/23/M/J/16/Q8

.....

.....

.....

.....

.....[3]

(b) State the equation that shows the decay of a particle in a nucleus that results in β^+ emission. All particles in the equation should be shown in the notation that is usually used for the representation of nuclides.

[2]

(c) (i) State the quark composition of

1. a proton,

.....

2. a neutron.

.....

[2]

(ii) Use the quark model to explain the charge on a proton.

.....

.....

.....[1]

27 (a) State **one** difference between a hadron and a lepton.

.....
[1]

(b) (i) State the quark composition of a proton and of a neutron.

proton:
 neutron:
 [2]

(ii) Use your answer in (i) to determine the quark composition of an α -particle.

quark composition:[1]

(c) The results of the α -particle scattering experiment provide evidence for the structure of the atom.

result 1: The vast majority of α -particles pass straight through the metal foil or are deviated by small angles.

result 2: A very small minority of α -particles are scattered through angles greater than 90° .

State what may be inferred from

(i) result 1,

.....
[1]

(ii) result 2.

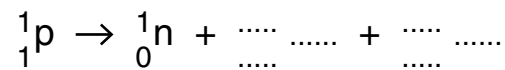
.....

[2]

28 (a) State **one** difference between a hadron and a lepton.

.....
[1]

(b) A proton within a nucleus decays to form a neutron and two other particles. A partial equation to represent this decay is



(i) Complete the equation. [2]

(ii) State the name of the interaction or force that gives rise to this decay.

.....[1]

(iii) State three quantities that are conserved in the decay.

1.

2.

3.

[3]

(c) Use the quark composition of a proton to show that it has a charge of $+e$, where e is the elementary charge.

Explain your working.

[3]

29 (a) Use the quark model to show that

9702/21/M/J/17/Q7

(i) the charge on a proton is $+e$,

.....[1]

(ii) the charge on a neutron is zero.

.....[1]

(b) A nucleus of $^{90}_{38}\text{Sr}$ decays by the emission of a β^- particle. A nucleus of $^{64}_{29}\text{Cu}$ decays by the emission of a β^+ particle.

(i) In Fig. 7.1, state the nucleon number and proton number for the nucleus produced in each of these decay processes.

	nucleus formed by β^- decay	nucleus formed by β^+ decay
nucleon number		
proton number		

Fig. 7.1

[1]

(ii) State the name of the force responsible for β decay.

.....[1]

(iii) State the names of the leptons produced in each of the decay processes.

β^- decay:

β^+ decay:[1]

9702/22/M/J/17/Q8

30 (a) Describe **two** differences between the decay of a nucleus that emits a β^- particle and the decay of a nucleus that emits a β^+ particle.

1.

.....

2.

.....

[2]

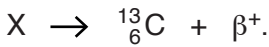
(b) In a simple quark model there are three types of quark. State the composition of the proton and of the neutron in terms of these three quarks.

proton:

neutron:

[1]

- 31** A stationary nucleus X decays by emitting a β^+ particle to form a nucleus of carbon-13 ($^{13}_6\text{C}$). An incomplete equation to represent this decay is



- (a)** State the name of the class (group) of particles that includes β^+ .
[1]
- (b)** For nucleus X, state the number of
 protons,
 neutrons.[1]
- (c)** The carbon-13 nucleus has a mass of $2.2 \times 10^{-26}\text{ kg}$. Its kinetic energy as a result of the decay process is 0.80 MeV .

Calculate the speed of this nucleus.

speed = ms^{-1} [3]

- (d)** Explain why the sum of the kinetic energies of the carbon-13 nucleus and the β^+ particle cannot be equal to the total energy released by the decay process.

[1]

- 32 (a) A nucleus X decays by emitting a β^+ particle to form a new nucleus, ${}^{23}_{11}\text{Na}$.

State the number of nucleons and the number of neutrons in nucleus X.

number of nucleons =

number of neutrons =

[2]

- (b) State one similarity and one difference between a β^+ particle and a β^- particle.

similarity:

difference:

[2]

- 33 A β^- particle from a radioactive source is travelling in a vacuum with kinetic energy 460 eV. The particle enters a uniform electric field at a right-angle and follows the path shown in Fig. 7.1.

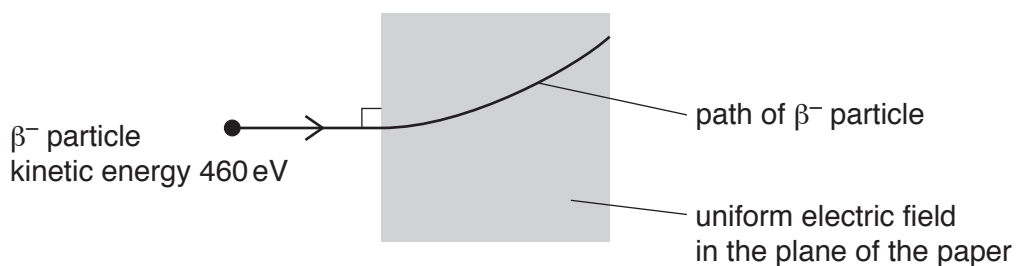


Fig. 7.1

- (a) The direction of the electric field is in the plane of the paper.
On Fig. 7.1, draw an arrow to show the direction of the electric field.

[1]

- (b) Calculate the speed of the β^- particle before it enters the electric field.

speed = ms^{-1} [3]

- (c) Other β^- particles from the same radioactive source travel outside the electric field along the same incident path as that shown in Fig. 7.1.

State and briefly explain whether those β^- particles will all follow the same path inside the electric field.

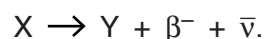
.....

.....

.....

.....[2]

- 34** A stationary nucleus X decays to form nucleus Y, as shown by the equation **9702/22/M/J/18/Q7**



- (a) In the above equation, draw a circle around all symbols that represent a lepton. [1]

- (b) State the name of the particle represented by the symbol $\bar{\nu}$.

.....[1]

- (c) Energy is released during the decay process. State the form of the energy that is gained by nucleus Y.

.....[1]

- (d) By comparing the compositions of X and Y, state and explain whether they are isotopes.

.....

.....

.....[2]

- (e) The quark composition of one nucleon in X is changed during the emission of a β^- particle. Describe this change to the quark composition.

.....

.....[1]

35 A graph of nucleon number A against proton number Z is shown in Fig. 7.1.

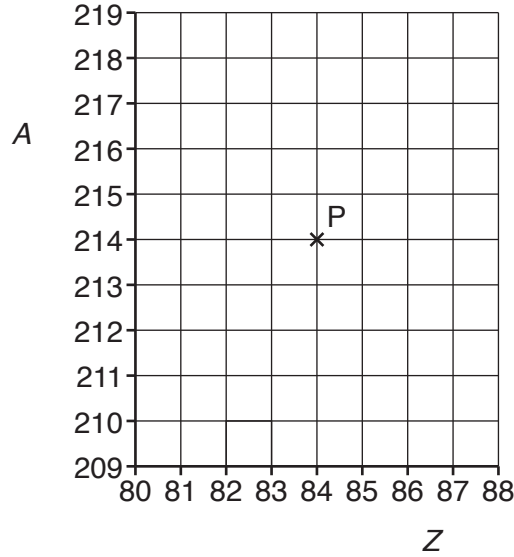


Fig. 7.1

The graph shows a cross (labelled P) that represents a nucleus P.

Nucleus P decays by emitting an α particle to form a nucleus Q.
Nucleus Q then decays by emitting a β^- particle to form a nucleus R.

- (a) On Fig. 7.1, use a cross to represent
- (i) nucleus Q (label this cross Q), [1]
 - (ii) nucleus R (label this cross R). [1]
- (b) State the name of the class (group) of particles that includes the β^- particle.
-[1]
- (c) The quark composition of one nucleon in Q is changed during the emission of the β^- particle. Describe this change to the quark composition.
-
-[1]
-

- 36 (a) State what is meant by an *electric field*.

.....
 [1]

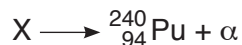
- (b) A particle of mass m and charge q is in a uniform electric field of strength E . The particle has acceleration a due to the field.

Show that

$$a = \frac{Eq}{m}.$$

[2]

- (c) A stationary nucleus X decays by emitting an α -particle to form a nucleus of plutonium, ${}^{240}_{94}\text{Pu}$, as shown.



- (i) Determine the number of protons and the number of neutrons in nucleus X.

number of protons =

number of neutrons =

[2]

- (ii) The total mass of the plutonium nucleus and the α -particle is less than that of nucleus X. Explain this difference in mass.

.....

 [2]

- (iii) The plutonium nucleus and the α -particle are both accelerated by the same uniform electric field.

Use the expression in (b) to determine the ratio

$$\frac{\text{acceleration of the } \alpha\text{-particle}}{\text{acceleration of the plutonium nucleus}} .$$

ratio = [2]

- 37 (a) In the following list, underline all particles that are leptons.

9702/22/O/N/18/Q8

antineutrino positron proton quark [1]

- (b) A stationary nucleus of magnesium-27, ${}^{27}_{12}\text{Mg}$, decays by emitting a β^- particle and γ radiation. An incomplete equation to represent this decay is



- (i) State the nucleon number and the proton number of nucleus X.

nucleon number =

proton number = [2]

- (ii) State the name of the interaction that gives rise to this decay.

.....[1]

- (iii) State **two** possible reasons why the sum of the kinetic energy of the β^- particle and the energy of the γ radiation is less than the total energy released during the decay of the magnesium nucleus.

1.
.....

2.
.....

[2]

- 38** A particle of mass m and charge q is in a uniform electric field of strength E . The particle has acceleration a due to the field.

(a) Show that

$$\frac{q}{m} = \frac{a}{E}.$$

[2]

- (b)** The particle has a charge of $4e$ where e is the elementary charge. The electric field strength is $3.5 \times 10^4 \text{ V m}^{-1}$. The acceleration of the particle is $1.5 \times 10^{12} \text{ m s}^{-2}$.

Use the expression in **(a)** to show that the mass of the particle is 9.0 u .

[2]

- (c)** The particle is a nucleus. State the number of protons and the number of neutrons in the nucleus.

number of protons =

number of neutrons =

[1]

- (d)** A second nucleus that is an isotope of the nucleus in **(c)** is in the same uniform electric field. State and explain whether the electric field produces, for the two nuclei, the same magnitudes of

(i) force,

.....

.....[1]

(ii) acceleration.

.....

.....[1]

- 39 (a) One of the results of the α -particle scattering experiment is that a very small minority of the α -particles are scattered through angles greater than 90° .

State what may be inferred about the structure of the atom from this result.

.....

.....

.....

.....[2]

- (b) A hadron has an overall charge of $+e$, where e is the elementary charge. The hadron contains three quarks. One of the quarks is a strange (s) quark.

- (i) State the charge, in terms of e , of the strange (s) quark.

charge = [1]

- (ii) The other two quarks in the hadron have the same charge as each other.

By considering charge, determine a possible type (flavour) of the other two quarks.
Explain your working.

.....

.....[2]

- 40 (a) State what is meant by a *field line* (*line of force*) in an electric field.

.....
[1]

- (b) An electric field has two different regions X and Y. The field strength in X is less than that in Y. Describe a difference between the pattern of field lines (lines of force) in X and in Y.

.....
[1]

- (c) A particle P has a mass of 0.15 u and a charge of $-1e$, where e is the elementary charge.

- (i) Particle P and an α -particle are in the same uniform electric field. Calculate the ratio

$$\frac{\text{magnitude of acceleration of particle P}}{\text{magnitude of acceleration of } \alpha\text{-particle}}.$$

ratio = [3]

- (ii) Particle P is a hadron composed of only two quarks. One of them is a down (d) quark.

By considering charge, determine a possible type (flavour) of the other quark.
 Explain your working.

.....
[3]

- 41** A sample of a radioactive substance may decay by the emission of either α -radiation or β -radiation and/or γ -radiation.

State the type of radiation, one in each case, that:

- (a) consists of leptons

.....[1]

- (b) contains quarks

.....[1]

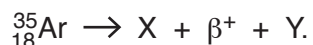
- (c) cannot be deflected by an electric field

.....[1]

- (d) has a continuous range of energies, rather than discrete values of energy.

.....[1]

- 42** (a) The decay of a nucleus ${}^{35}_{18}\text{Ar}$ by β^+ emission is represented by



A nucleus X and two particles, β^+ and Y, are produced by the decay.

State:

- (i) the proton number and the nucleon number of nucleus X

proton number =

nucleon number =

[1]

- (ii) the name of the particle represented by the symbol Y.

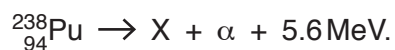
..... [1]

- (b) A hadron consists of two down quarks and one strange quark.

Determine, in terms of the elementary charge e , the charge of this hadron.

charge = [2]

- 43** A nucleus of plutonium-238 ($^{238}_{94}\text{Pu}$) decays by emitting an α -particle to produce a new nucleus X and 5.6 MeV of energy. The decay is represented by



- (a)** Determine the number of protons and the number of neutrons in nucleus X.

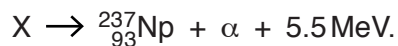
number of protons =

number of neutrons = [2]

- (b)** Calculate the number of plutonium-238 nuclei that must decay in a time of 1.0 s to produce a power of 0.15 W.

number = [2]

- 44 A stationary nucleus of a radioactive isotope X decays by emitting an α -particle to produce a nucleus of neptunium-237 and 5.5 MeV of energy. The decay is represented by



- (a) Calculate the number of protons and the number of neutrons in a nucleus of X.

number of protons =

number of neutrons = [2]

- (b) Explain why the energy transferred to the α -particle as kinetic energy is less than the 5.5 MeV of energy released in the decay process.

.....
 [1]

- (c) A sample of X is used to produce a beam of α -particles in a vacuum. The number of α -particles passing a fixed point in the beam in a time of 30 s is 6.9×10^{11} .

- (i) Calculate the average current produced by the beam of α -particles.

current = A [2]

- (ii) Determine the total power, in W, that is produced by the decay of 6.9×10^{11} nuclei of X in a time of 30 s.

power = W [2]

45. 9702/22/F/M/21/Q7

- (a) The results of the α -particle scattering experiment provide evidence for the structure of the atom.

Result 1: The vast majority of the α -particles pass straight through the metal foil or are deviated by small angles.

Result 2: A very small minority of α -particles is scattered through angles greater than 90° .

State what may be inferred (deduced) from:

- (i) result 1

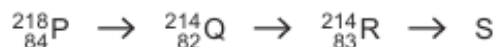
.....
 [1]

- (ii) result 2.

.....

 [2]

- (b) A radioactive decay sequence contains four nuclei, P, Q, R and S, as shown.



Nucleus S is an isotope of nucleus P.

- (i) Determine the proton number and the nucleon number of nucleus S.

proton number =

nucleon number =

[2]

- (ii) The quark composition of a nucleon in Q changes as Q decays to form R.

Describe this change to the quark composition of the nucleon.

.....
 [1]

[Total: 6]

46. 9702/22/M/J/21/Q6

- (a) One of the results of the α -particle scattering experiment is that a very small minority of the α -particles are scattered through angles greater than 90° .

State what may be inferred about the structure of the atom from this result.

.....
.....
.....
..... [2]

- (b) An α -particle is made up of other particles. One of these particles is a proton.

State and explain whether a proton is a fundamental particle.

.....
..... [1]

- (c) A radioactive source produces a beam of α -particles in a vacuum. The average current produced by the beam is $6.9 \times 10^{-9} \text{ A}$.

Calculate the average number of α -particles passing a fixed point in the beam in a time of 1.0 minute.

number = [3]

47. 9702/23/M/J/21/Q6

(a) State the quark composition of:

(i) a proton

..... [1]

(ii) a neutron

..... [1]

(iii) an alpha-particle.

.....
..... [2]

(b) In the alpha-particle scattering experiment, alpha-particles were directed at a thin gold foil.

State what may be inferred from:

(i) the observation that most alpha-particles pass through the foil

..... [1]

(ii) the observation that some alpha-particles are scattered through angles greater than 90° .

.....
.....
..... [2]

48. 9702/21/O/N/21/Q6

- (a) Complete Table 6.1 to show the masses (in terms of the unified atomic mass unit u) and charges (in terms of the elementary charge e) of α , β^+ and β^- particles.

Table 6.1

	mass/ u	charge/ e
α -particle		
β^+ particle		
β^- particle		

[4]

- (b) Carbon-14 is radioactive and decays by emission of β^- particles.

- (i) Nuclei do not contain β^- particles.

Explain the origin of the β^- particle that is emitted from the nucleus during β^- decay.

.....

 [1]

- (ii) State the change in the quark composition of a carbon-14 nucleus when it emits a β^- particle.

..... [1]

- (iii) Suggest why the β^- particles are emitted with a range of different energies.

.....

 [2]

[Total: 8]

49. 9702/22/O/N/21/Q7

A stationary nucleus P of mass 243u decays by emitting an α -particle of mass 4u to form a different nucleus Q, as illustrated in Fig. 7.1.

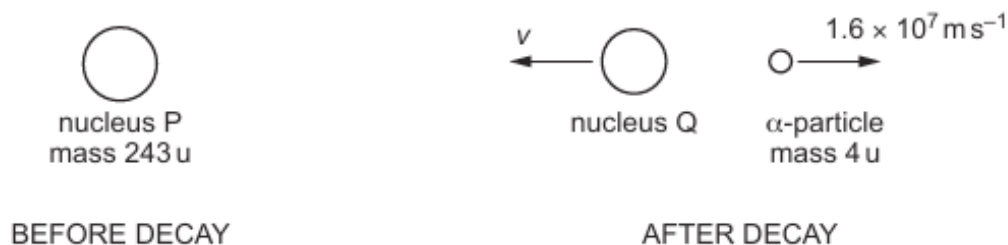


Fig. 7.1

The initial speed of the α -particle is $1.6 \times 10^7 \text{ m s}^{-1}$.

- (a) Use the principle of conservation of momentum to explain why the initial velocities of nucleus Q and the α -particle must be in opposite directions.

.....

 [2]

- (b) Determine the initial speed v of nucleus Q.

$v = \dots\dots\dots \text{ m s}^{-1}$ [2]

- (c) Calculate the initial kinetic energy, in MeV, of the α -particle.

kinetic energy = $\dots\dots\dots \text{ MeV}$ [3]

(d) A graph of number of neutrons N against proton number Z is shown in Fig. 7.2.

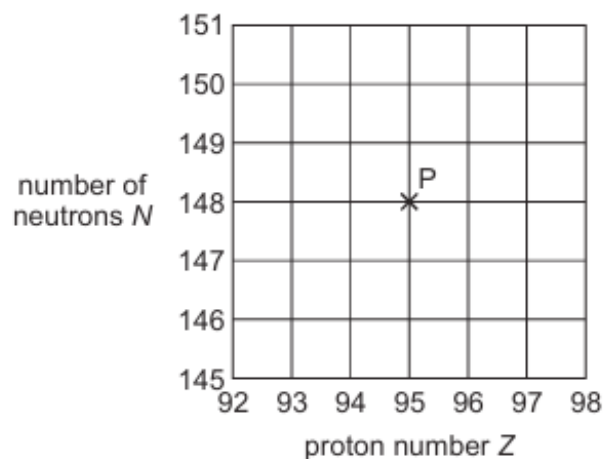


Fig. 7.2

The graph shows a cross that represents nucleus P.

A nucleus R has a nucleon number of 242 and is an isotope of nucleus P.

Nucleus R decays by emitting a β^- particle to form a different nucleus S.

(i) On Fig. 7.2, draw a cross to represent:

1. nucleus R (label this cross R)
2. nucleus S (label this cross S).

[2]

(ii) State the name of the other lepton, in addition to the β^- particle, that is emitted during the decay of nucleus R.

..... [1]

[Total: 10]

50. 9702/22/F/M/22/Q7

- (a) A nucleus of sodium-22 ($^{22}_{11}\text{Na}$) decays by emitting a β^+ particle. A different nucleus is formed by the decay.

- (i) State the name of another lepton that is produced by the decay.

..... [1]

- (ii) Determine the nucleon number and the proton number of the nucleus that is formed by the decay.

nucleon number =

proton number =

[2]

- (iii) The quark composition of a nucleon in the sodium-22 nucleus is changed during the decay.

Describe the change to the quark composition of the nucleon.

.....

..... [1]

- (b) A baryon consists of quarks that are the same flavour (type). The charge of the baryon is $-2e$, where e is the elementary charge.

- (i) Calculate, in terms of e , the charge of each quark.

charge = e [1]

- (ii) State a possible flavour (type) of the quarks.

..... [1]

[Total: 6]

51. 9702/21/M/J/22/Q7

(a) An unstable nucleus ${}^A_Z\text{X}$ decays by emitting a β^- particle.

(i) Determine quantitatively the changes, if any, in A and Z when X decays.

change in A =

change in Z =

[2]

(ii) In addition to the β^- particle, another lepton is emitted during the decay.

State the name of the other lepton that is emitted.

..... [1]

(b) A particle P is composed of an up quark (u) and a down antiquark ($\bar{d}$).

(i) Calculate the charge q of particle P in terms of e , where e is the elementary charge.

Show your working.

$q = \dots\dots\dots e$ [2]

(ii) Particle P belongs to **two** classes (groups) of particles.

State the names of these two classes.

1

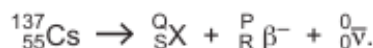
2

[2]

[Total: 7]

52. 9702/22/M/J/22/Q7

- (a) A nucleus of caesium-137 ($^{137}_{55}\text{Cs}$) decays by emitting a β^- particle to produce a nucleus of an element X and an antineutrino. The decay is represented by



- (i) State the number represented by each of the following letters.

P

Q

R

S

[2]

- (ii) State the name of the class (group) of particles that includes the β^- particle and the antineutrino.

..... [1]

- (b) A particle Y has a quark composition of ddd where d represents a down quark.

A particle Z has a quark composition of $\bar{u}d$ where $\bar{u}$ represents an up antiquark.

- (i) Show that the charges of particles Y and Z are equal.

[2]

- (ii) State and explain which particle is a meson and which particle is a baryon.

meson:

.....

baryon:

.....

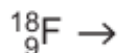
[2]

[Total: 7]

53. 9702/23/M/J/22/Q7

- (a) Fluorine-18 ($^{18}_9\text{F}$) is an isotope that decays to an isotope of oxygen (O) by the emission of a β^+ particle.

- (i) Complete the nuclear equation for the decay, including all the particles involved.



[3]

- (ii) A quark in the fluorine-18 nucleus changes flavour during the decay. State this change of flavour.

..... quark to quark.

[1]

- (b) A hadron has a charge of $-2e$, where e is the elementary charge.

- (i) State and explain whether the hadron is a meson or a baryon.

.....
.....
..... [2]

- (ii) State a possible quark composition for the hadron.

.....
..... [1]

[Total: 7]

54. 9702/21/O/N/22/Q6

- (a)** A lepton is an example of a fundamental particle.

State what is meant by fundamental particle.

.....

..... [1]

- (b)** A lambda particle Λ^0 is a hadron that consists of an up (u) quark, a down (d) quark and a strange (s) quark.

Show that the charge on the Λ^0 particle is zero.

[2]

- (c)** The Λ^0 particle is unstable. It can decay into a neutron (n) and a pion (π^0) as shown by

$$\Lambda^0 \rightarrow n + \pi^0.$$

The π^0 particle consists of an up quark and an up antiquark.

- (i)** Compare the properties of an up quark and an up antiquark.

.....

.....

..... [2]

- (ii)** Explain why the neutron is classed as a baryon and the π^0 particle is classed as a meson.

.....

.....

..... [2]

[Total: 7]

55. 9702/22/O/N/22/Q7

- (a) Describe the structure of an **atom** of uranium-238, $^{238}_{92}\text{U}$.

.....

 [2]

- (b) The decay of uranium-238 is shown by the equation



For nucleus X, calculate the ratio, in C kg^{-1} , of

$$\frac{\text{charge}}{\text{mass}}.$$

ratio = C kg^{-1} [3]

- (c) Two particles P and Q each consist of three quarks. These quarks are up (u) or down (d) quarks.

Particle P has no overall charge.

Particle Q has an overall charge of $+2e$, where e is the elementary charge.

State the quark composition of:

- (i) particle P

..... [1]

- (ii) particle Q.

..... [1]

[Total: 7]

56. 9702/23/O/N/22/Q6

- (a) The nuclide $^{14}_6\text{C}$ (carbon-14) is unstable and undergoes β^- decay, emitting a high-energy electron and an antineutrino to form a new nuclide X. The equation for this decay is shown.



Complete the equation.

[2]

- (b) (i) State the equation for β^- decay in terms of the fundamental particles involved.

[1]

- (ii) Use your equation from (b)(i) to show how charge is conserved in β^- decay.

[1]

- (c) Neutrinos were first proposed to exist more than 20 years before they were directly detected, in order to explain a particular experimental observation about β -decay.

- (i) State an observation about β -decay that is explained by the existence of neutrinos.

.....

 [1]

- (ii) Suggest how the existence of neutrinos explains the observation in (c)(i).

.....

 [1]

[Total: 6]

57. 9702/22/F/M/23/Q7

- (a)** Nuclei X and Y are different isotopes of the same element.

Nucleus X is unstable and emits a β^+ particle to form nucleus Z.

By comparing the number of protons in each nucleus, state and explain whether the charge of nucleus X is less than, the same as or greater than the charge of:

- (i)** nucleus Y

.....
..... [1]

- (ii)** nucleus Z.

.....
.....
..... [2]

- (b)** Hadrons can be divided into two groups (classes), P and Q. Group P is baryons.

- (i)** State the name of group Q.

..... [1]

- (ii)** Describe, in general terms, the quark structure of hadrons that belong to group Q.

.....
..... [1]

[Total: 5]

58. 9702/21/M/J/23/Q8

An isolated stationary nucleus X decays by emitting an α -particle to form a nucleus Y.

Nucleus Y and nucleus Z are isotopes of the same element.

- (a) By comparing the number of protons in each nucleus, state and explain whether the charge of nucleus Y is less than, greater than or the same as the charge of:

- (i) nucleus Z

.....
..... [1]

- (ii) nucleus X.

.....
.....
..... [2]

- (b) Use the principle of conservation of momentum to explain why nucleus Y cannot be stationary immediately after the decay of nucleus X.

.....
.....
.....
..... [2]

[Total: 5]

59. 9702/22/M/J/23/Q8

- (a)** Nucleus P and nucleus Q are isotopes of the same element.

Nucleus Q is unstable and emits a β^- particle to form nucleus R.

- (i)** For nuclei P and Q, compare:

- the number of protons

.....

- the number of neutrons.

.....

[2]

- (ii)** When nucleus Q decays to form nucleus R, the quark composition of a nucleon changes.

State the change to the quark composition of the nucleon.

..... [1]

- (iii)** State the name of another particle that must be emitted from nucleus Q in addition to the β^- particle.

..... [1]

- (b)** A hadron consists of two charm quarks and one bottom quark.

Determine, in terms of the elementary charge e , the charge of the hadron.

charge = e [2]

[Total: 6]

60. 9702/23/M/J/23/Q7

- (a) Table 7.1 shows incomplete data for three flavours (types) of quark. The elementary charge is e .

Table 7.1

flavour	quark		antiquark	
	symbol	charge / e	symbol	charge / e
up	u	$+\frac{2}{3}$	$\bar{u}$	
down	d		$\bar{d}$	
charm	c		$\bar{c}$	

Complete Table 7.1 by inserting the missing charges. [2]

- (b) Using the symbols given in Table 7.1, state a possible quark combination for the following hadrons:

- (i) a neutral baryon

..... [1]

- (ii) a meson with a charge of $+e$.

..... [1]

- (c) Quarks are fundamental particles.

Electrons are in another group (class) of fundamental particle.

- (i) State the name of this group.

..... [1]

- (ii) State the name of another particle in this group.

..... [1]

[Total: 6]

61. 9702/21/O/N/23/Q7

- (a) The results of the α -particle scattering experiment led to the development of the nuclear model for the atom.

State the results that suggested that most of the mass of the atom is concentrated in a very small region and most of the atom is empty space.

.....

.....

.....

..... [2]

- (b) State the composition of γ -radiation.

..... [1]

- (c) Table 7.1 lists the names of three particles and possible classifications for them.

Table 7.1

particle name	classification		
	baryon	hadron	lepton
neutrino			
neutron			
positron			

Complete Table 7.1 by placing ticks (✓) in the boxes to indicate the classifications that apply to each particle. [2]

- (d) The discovery of a particle with an unusual charge was an important step in the development of the theory of quarks. The particle is a hadron with a mass of 2.19×10^{-27} kg and a charge of $+2e$, where e is the elementary charge.

- (i) Calculate the mass, in u, of the particle. Give your answer to three significant figures.

mass = u [1]

- (ii) Determine a possible quark composition of a hadron with a charge of $+2e$.
Explain your reasoning.

[2]

[Total: 8]

62. 9702/22/O/N/23/Q7

(a) In the following list, underline **all** the particles that are **not** fundamental.

antineutrino baryon nucleon positron [1]

(b) A nucleus of thorium-230 ($^{230}_{90}\text{Th}$) decays in stages, by emitting α -particles and β^- particles, to form a nucleus of lead-206 ($^{206}_{82}\text{Pb}$).

Determine the total number of α -particles and the total number of β^- particles that are emitted during the sequence of decays that form the nucleus of lead-206 from the nucleus of thorium-230.

number of α -particles =

number of β^- particles =

[2]

(c) A meson has a charge of $-1e$, where e is the elementary charge. The quark composition of the meson includes a charm antiquark.

State and explain a possible flavour (type) of the other quark in the meson.

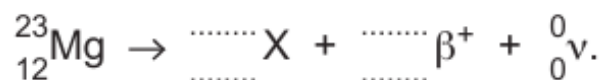
.....

..... [2]

[Total: 5]

63.9702/23/O/N/23/Q8

- (a) The nuclide ${}^{23}_{12}\text{Mg}$ is an isotope of magnesium that undergoes β^+ decay to form a new nuclide X according to the equation



Four numbers are missing from the equation.

- (i) For the nuclide ${}^{23}_{12}\text{Mg}$, state what is represented by the numbers 23 and 12.

23 represents:

12 represents: [2]

- (ii) Complete the equation by inserting the missing numbers. [2]

- (iii) State the name of the group (class) of fundamental particles to which the positron and neutrino belong.

..... [1]

- (b) A radioactive source emits particles from its nuclei when it decays.

Fig. 8.1 shows, for the source, the variation with kinetic energy of the number of particles emitted.

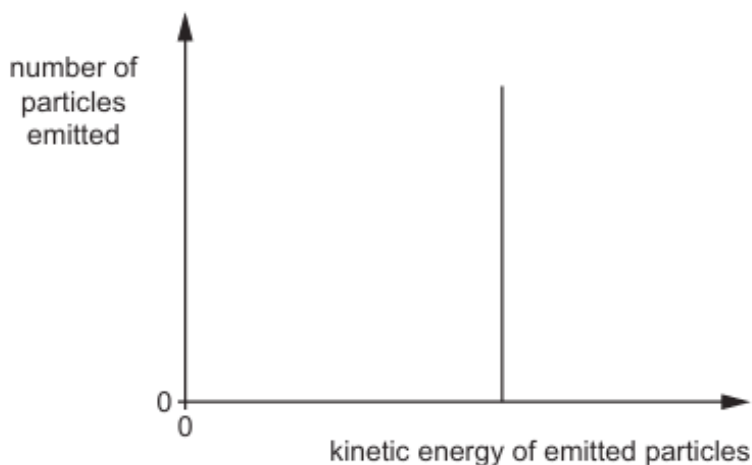


Fig. 8.1

State how Fig. 8.1 shows that these nuclei do **not** undergo beta-decay.

.....

..... [1]

[Total: 6]

ANSWERS

1.

- (a) rate of decay / activity / decay (of nucleus) is not affected by external factors / environment / surroundings B2
(if states specific factor(s), rather than giving general statement above, then give 2 marks for two stated factors, but 1 mark only if one factor stated)
- (b) (i) gamma / γ B1
 (ii) alpha / α B1
 (iii) gamma / γ B1
 (iv) beta / β B1

2.

- (a) deviation shown correctly B1
- (b) smaller deviation (not zero deviation) M1
 acceptable path wrt position of N A1
- (c) the nucleus is (very) small M1
 in comparison to the atom A1
(special case: 'atom is mostly empty space' scores 1 mark)
- (d) deviation depends on charge on the nucleus / N / electrostatic repulsion B1
 same charge so no change in deviation B1

3.

- (a) *either* forms of same element
or atoms / nuclei with same number of protons M1
 atoms / nuclei contain different numbers of neutrons A1
(use of 'element' rather than atoms / nuclei scores max 1 mark)
- (b) (i) decay is not affected by environmental factors B1
(allow two named factors)
- (ii) *either* time of decay (of a nucleus) cannot be predicted
or nucleus has constant probability in a given time B1
- (c) $^{185}_{75}\text{Re}$ B1
either $^0_{-1}\text{e}$ *or* $^0_{-1}\beta$ B1

- 4.
- (a) nuclei/atoms with same proton number/atomic number B1
 nuclei/atoms contain different numbers of neutrons/different atomic mass B1
- (b) (i) 92 A1
 (ii) 146 A1
- (c) (i) mass = $238 \times 1.66 \times 10^{-27}$ C1
 = 3.95×10^{-25} kg A1
- (ii) volume = $\frac{4}{3} \pi \times (8.9 \times 10^{-15})^3$ (= 2.95×10^{-42}) C1
 density = $(3.95 \times 10^{-25}) / (2.95 \times 10^{-42})$
 = 1.3×10^{17} kg m⁻³ A1
- (d) nucleus contains most of mass of atom B1
 either nuclear diameter/volume very much less than that of atom
 or atom is mostly (empty) space B1
- 5.
- (a) (i) *either* helium nucleus
 or contains 2 protons and 2 neutrons B1
- (ii) e.g. range is a few cm in air/sheet of thin paper
 speed up to 0.1 c
 causes dense ionisation in air
 positively charged or deflected in magnetic or electric fields
 (any two, 1 each to max 2) B2
- (b) (i) ${}^4_2\alpha$ B1
 either 1_1p or 1_1H B1
- (ii) 1 initially, α -particle must have some kinetic energy B1
- (ii) 2 1.1 MeV = $1.1 \times 1.6 \times 10^{-13} = 1.76 \times 10^{-13}$ J C1
 $E_K = \frac{1}{2}mv^2$ C1
 $1.76 \times 10^{-13} = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$ C1
 $v = 7.3 \times 10^6$ m s⁻¹ A1
 use of 1.67×10^{-27} kg for mass is a maximum of 3/4

- 6.
- (a) *either* different forms of same element
or nuclei have same number of protons
different numbers of neutrons (in the nucleus) M1
A1
- (b) (i) proton number conserved B1
nucleon number conserved B1
mass-energy conserved B1
- (ii) 1. $Z = 36$ A1
2. $x = 3$ A1
- 7.
- (a) (i) most α -particles were deviated through small angles B2
(*allow 1 mark for 'straight through' / undeviated*)
- (ii) small fraction of α -particles deviated through large angles M'
greater than 90° (*allow rebound back*) A1
- (b) e.g. β -particles have a range of energies
 β -particles deviated by (orbital) electrons
 β -particle has (very) small mass
(*any two sensible suggestions, 1 each, max 2*) B2
- Do not allow β -particles have negative charge or β -particles have high speed
- 8.
- (a) nuclei with the same number of protons B1
and a different number of neutrons B1
- (b) (i) (mass + energy) (taken together) is conserved (B1)
momentum is conserved (B1)
one point required max. 1 B1
- (ii) $a = 1$ and $b = 0$ B1
 $x = 56$ B1
 $y = 92$ B1
- (c) proton number = 90 B1
nucleon number = 235 B1

9.

- (a) (i) the half life / count rate / rate of decay / activity is the same no matter what external factors / environmental factors **or** two named factors such as temperature and pressure changes are applied B1
- (ii) the observations of the count rate / count rate / rate of decay / activity / radioactivity during decay shows variations / fluctuations B1

(b)

property	α -particle	β -particle	γ -radiation
charge	(+)2e	-e	0
mass	4u	$9.11 \times 10^{-31} \text{ kg}$	0
speed	0.01 to 0.1 c	up to 0.99 c	c

one mark for each correct line

B3

- (c) collision with molecules B1
causes ionisation (of the molecule) / electron is removed B1

10.

- (a) (i) 2 protons and 2 neutrons B1
- (ii) e.g. positively charged 2e
mass 4u
constant energy
absorbed by thin paper *or* few cm of air (3 cm $\rightarrow$ 8 cm)
(*not low penetration*)
highly ionizing
deflected in electric/magnetic fields
(*One mark for each property, max 2*) B2
- (b) mass-energy is conserved B1
difference in mass 'changed' into a form of energy B1
energy in the form of kinetic energy of the products / γ -radiation
photons / e.m. radiation B1

11.

- (a) W = 1 and X = 0 A1
Y = 2 A1
Z = 55 A1
- (b) explanation in terms of mass – energy conservation B1
energy released as gamma or photons or kinetic energy of products or
em radiation B1

12.

- (a) thin paper reduces count rate hence α B1
 addition of 1 cm of aluminium causes little more count rate reduction hence only other radiation is γ B1
- (b) magnetic field perpendicular to direction of radiation B1
 look for a count rate in expected direction / area if there were negatively charged radiation present. If no count rate recorded then β not present. B1

13.

- (a) 92 protons in the nucleus and 92 electrons around nucleus B1
 143 neutrons (in the nucleus) B1
- (b) (i) α -particle travels short distance in air B1
 (ii) very small proportion in backwards direction / large angles B1
 majority pass through with no /small deflections B1
 either most of mass is in very small volume (nucleus) and is charged or most of atom empty space B1
- (c) $I = Q / t$ C1
 $n/t = (1.5 \times 10^{-12}) / (2 \times 1.6 \times 10^{-19})$ C1
 $n/t = 4.7 \times 10^6 \text{ s}^{-1}$ A1

14.

- (a) ${}^3_2\text{He} + {}^3_2\text{He} \rightarrow {}^4_2\text{He} + 2 {}^1_1\text{p} + Q$
 A numbers correct (4 and 1) B1
 Z numbers correct (2 and 1) B1
- (b) both nuclei have 2 protons B1
 the two isotopes have 1 neutron and two neutrons B1
 [allow 1 for 'same number of protons but different number of neutrons']
- (c) proton number and neutron number B1
 energy – mass B1
 momentum B1
- (d) (i) γ radiation B1
 (ii) product(s) must have kinetic energy B1
- (e) $13.8 \text{ MeV} = 13.8 \times 1.6 \times 10^{-19} \times 10^6 (= 2.208 \times 10^{-12})$ C1
 $60 = n \times 13.8 \times 1.6 \times 10^{-13}$
 $n = 2.7(2) \times 10^{13} \text{ s}^{-1}$ A1

15.

- (a) (i) electron B1
 (ii) any **two**:
 can be deflected by electric and magnetic fields or negatively charged /
 absorbed by few (1 – 4) mm of aluminum / 0.5 to 2 m or metres for range in air /
 speed up to 0.99c / range of speeds / energies B2
 (iii) decay occurs and cannot be affected by external / environmental factors
 or two stated factors such as chemical / pressure / temperature / humidity B1
- (b) 3 and 0 for superscript numbers B1
 2 and –1 for subscript numbers B1
- (c) energy = $5.7 \times 10^3 \times 1.6 \times 10^{-19}$ (= 9.12×10^{-16} J) C1

$$v^2 = \frac{2 \times 9.12 \times 10^{-16}}{9.11 \times 10^{-31}}$$
 C1

$$v = 4.5 \times 10^7 \text{ ms}^{-1}$$
 A1
- (d) both have 1 proton and 1 electron B1
 1 neutron in hydrogen-2 and 2 neutrons in hydrogen-3 B1
 (special case: for one mark 'same number of protons / atomic number
 different number of neutrons')

16.

- (a) the majority/most went straight through
 or were deviated by small angles B1
 a very small proportion/a few were deviated by large angles B1
 small angles described as $< 10^\circ$ and large angles described as $> 90^\circ$ B1
- (b) most of the atom is empty space/nucleus very small compared with atom B1
 mass and charge concentrated in (very small) nucleus B1
 correct links made with statements in (a) B1

17.

- (a) (i) $W = 206$ and $X = 82$
 $Y = 4$ and $Z = 2$ A1
- (ii) mass-energy is conserved A1
 mass on rhs is less because energy is released B1
- (b) not affected by external conditions/factors/environment B1
 or two examples temperature and pressure B1

18.

- (a) (i) the direction of the fields is the same OR fields are uniform OR constant electric field strength OR $E = V / d$ with symbols explained B1
- (ii) reduce p.d. across plates B1
increase separation of plates B1
- (iii) α opposite charge to β (as deflection in opposite direction) B1
 β has a range of velocities OR energies (as different deflections) and
 α all have same velocity OR energy (as constant deflection) B1
 α are more massive (as deflection is less for greater field strength) B1
- (b) $W = 234$ and $X = 90$ B1
 $Y = 4$ and $Z = 2$ B1
- (c) $A = 32$ and $B = 16$ and $C = 0$ and $D = -1$ B1

19.

- (a) α : helium nucleus
 β : electron
 γ : electromagnetic radiation / wave / ray or photon
three correct 2/2, two correct 1/2 B2
- (b) (i) atomic number / proton number / $Z - 2$, nucleon / mass number / $A - 4$ B1
- (ii) atomic number / proton number / $Z + 1$ B1
nucleon / mass number / A no change B1
- (iii) no change in proton or mass number or "no change" B1

20.

- (a) (i) A: 206, nucleon(s) or neutron(s) and proton(s) }
B: 82, proton(s) } all correct A1
- (ii) kinetic / E_K / KE B1
- (b) energy = $5.3 \times 1.6 \times 10^{-13}$ (J) [= 8.48×10^{-3} (J)] C1
power = $(7.1 \times 10^{18} \times 5.3 \times 1.6 \times 10^{-13}) / (3600 \times 24)$ = 70 (69.7) W

21.

(a) 92 protons and 143 neutrons

B1

(b)

	value
a	1
b	0
c	141
d	55

(a and b both required)

B1

B1

B1

(c) kinetic energy (of products) or gamma/ γ (radiation or photon)

B1

(d) (total) mass on left-hand side/reactants is greater than (total) mass on right-hand side/products

M1

difference in mass is (converted to) energy

A1

22.

(a) (i) (rate of decay) not affected by any external factors **or** changes in temperature and pressure etc.

B1

(ii) two protons and two neutrons

B1

(b) (i) (total) mass before decay/on left-hand side is greater than (total) mass on right-hand side/after the decay

M'

the difference in mass is released as kinetic energy of the products

A1

(may also be some γ radiation) (to conserve mass-energy)

(ii) $(6.2 \times 10^6 \times 1.6 \times 10^{-19} =) 9.9(2) \times 10^{-13} \text{ J}$

A1

23.

(a) ${}^1_1\text{p}$

B1

${}^0_{-1}\beta^-$ and ${}^0_0\bar{\nu}$

B1

(b) an (electron) antineutrino

B1

(c) lepton(s)

B1

(d) (i) down, down, up/ddu

B1

(ii) a down/d (quark) changes to an up/u (quark) **or** ddu $\rightarrow$ uud

B1

24.

- (a) hadron: neutron/proton
and
lepton: electron/(electron) neutrino
(allow other correct particles) B1
- (b) (i) proton: up up down or uud B1
(ii) neutron: up down down or udd B1
- (c) (i) neutron $\rightarrow$ proton + electron + (electron) antineutrino B1
(ii) up down down (quarks) change to up up down (quarks)
or
down (quark) changes to up (quark) B1

25.

- (a) both electron and neutrino: lepton(s) B1
both neutron and proton: hadron(s)/baryon(s) B1
- (b) (i) ${}^1_1\text{p} \rightarrow {}^1_0\text{n} + {}^0_1\beta + {}^0_0\nu$
correct symbols for particles M1
correct numerical values (allow no values on neutrino) A1
(ii) up up down or uud $\rightarrow$ up down down or udd B1
(iii) weak (nuclear) B1

26.

- (a) α -particle is 2 protons and 2 neutrons; β^- -particle is positive electron/positron
 α -particle has charge +2e; β^+ -particle has +e charge
 α -particle has mass 4u; β -particle has mass (1/2000)u
 α -particle made up of hadrons; β^+ -particle a lepton
any three B3
- (b) ${}^1_1\text{p} \rightarrow {}^1_0\text{n} + {}^0_1\beta + {}^0_0\nu$
all terms correct M1
all numerical values correct (ignore missing values on ν) A1
- (c) (i) 1. proton: up, up, down / uud B1
2. neutron: up, down, down / udd B1
(ii) up quark has charge +2/3 (e) and down quark has charge -1/3 (e)
total is +1(e) B1

27.

- (a) hadron not a fundamental particle/lepton is fundamental particle
 or
 hadron made of quarks/lepton not made of quarks
 or
 strong force/interaction acts on hadrons/does not act on leptons B1
- (b) (i) proton: up, up, down / uud B1
 neutron: up, down, down / udd B1
- (ii) composition: $2(\text{uud}) + 2(\text{udd})$
 $= 6 \text{ up, } 6 \text{ down} / 6\text{u, } 6\text{d}$ B1
- (c) (i) most of the atom is empty space
 or
 the nucleus (volume) is (very) small compared to the atom B1
- (ii) nucleus is (positively) charged B1
 the mass is concentrated in (very small) nucleus/small region/small volume/small core
 or
 the majority of mass in (very small) nucleus/small region/small volume/small core B1

28.

- (a) hadron not a fundamental particle/lepton is fundamental particle
 or hadron made of quarks/lepton not made of quarks
 or strong force/interaction acts on hadrons/does not act on leptons B1
- (b) (i) ${}^0_1\text{e}^{(+)}$ or ${}^0_1\beta^{(+)}$ B1
 ${}^0_0\nu_{(\text{e})}$ B1
- (ii) weak (nuclear force / interaction) B1
- (iii) • mass-energy
 • momentum
 • proton number
 • nucleon number
 • charge
Any three of the above quantities, 1 mark each B3
- (c) (quark structure of proton is) up, up, down or uud B1
 up/u (quark charge) is $(+) \frac{2}{3}(e)$, down/d (quark charge) is $- \frac{1}{3}(e)$ C1
 $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = (+)e$ A1

29.

- (a) (i) (proton is uud so) $(2/3)e + (2/3)e - (1/3)e = e$
 (a) (ii) (neutron is udd so) $(2/3)e - (1/3)e - (1/3)e = 0$

B1

B1

B1

(b)(i)

	β^-	β^+
nucleon number	90	64
proton number	39	28

all correct

- (ii) weak (nuclear force/interaction)

B1

- (iii) β^- decay: electron and (electron) antineutrino

B1

β^+ decay: positron and (electron) neutrino

30.

Question	Answer	Marks
8(a)	β^- emission: neutron changes to proton (+ beta ⁻ /electron) and β^+ emission: proton changes to neutron (+ beta ⁺ /positron)	B1
	β^- emission: (electron) antineutrino also emitted and β^+ emission: (electron) neutrino also emitted	B1
8(b)	proton: up up down (and zero strange) neutron: up down down (and zero strange)	B1

31.

7(a)	lepton(s)	B1
7(b)	protons: 7 and neutrons: 6	A1
7(c)	$E = \frac{1}{2}mv^2$	C1
	$= 0.80 \times 10^6 \times 1.60 \times 10^{-19}$	C1
	$= 1.28 \times 10^{-13} \text{ (J)}$	A1
	$v^2 = 2 \times 1.28 \times 10^{-13} / 2.2 \times 10^{-26}$ $v = 3.4 \times 10^6 \text{ m s}^{-1}$	
7(d)	an (electron) neutrino/ $\nu_{(e)}$ is also produced (and this has energy)	B1

32.

7(a)	nucleons = 23	B1
	neutrons = 11	B1
7(b)	similarity: same (rest) mass or equal (magnitude of) charge	B1
	difference: opposite (sign of) charge or one is matter and one is antimatter or one is an electron and one is an antielectron	B1

33.

7(a)	arrow pointing vertically down the page	B1
7(b)	$E = \frac{1}{2}mv^2$	C1
	$E = 460 \times 1.60 \times 10^{-19} (= 7.36 \times 10^{-17} \text{ (J)})$	C1
	$v = [(2 \times 460 \times 1.60 \times 10^{-19}) / (9.11 \times 10^{-31})]^{1/2}$ $= 1.3 \times 10^7 \text{ ms}^{-1}$	A1
7(c)	β^- particles have range of/different/various speeds/velocities/momenta/energies	M1
	so they follow different paths	A1

34.

7(a)	circle(s) drawn only around β^- and $\bar{\nu}$ symbols	B1
7(b)	(electron) <u>antineutrino</u>	B1
7(c)	kinetic (energy)	B1
7(d)	Y has one more proton (and one less neutron)/X has one less proton (and one more neutron) or Y has more protons (and fewer neutrons)/X has fewer protons (and more neutrons) or a neutron changes to a proton or the number of protons increases	M1
	(so) not isotopes	A1
7(e)	up down down changes to up up down or udd $\rightarrow$ uud or down changes to up or d $\rightarrow$ u	B1

35.

7(a)(i)	Q plotted at (82, 210)	A1
7(a)(ii)	R plotted at (83, 210)	A1
7(b)	lepton(s)	B1
7(c)	up down down changes to up up down or udd $\rightarrow$ uud or down changes to up or d $\rightarrow$ u	B1

36.

5(a)	region (of space) where a force acts on a (stationary) charge	B1
5(b)	$E = F/Q$	B1
	$F = ma$ and (so) $a = \frac{Eq}{m}$	A1
5(c)(i)	protons = 96	A1
	neutrons = 148	A1
5(c)(ii)	mass-energy is conserved/mass change is 'seen' as energy	B1
	energy released as gamma (radiation)/KE of α /KE of Pu	B1
5(c)(iii)	ratio = $\frac{2}{4} \times \frac{240}{94}$ or ratio = $\frac{2 \times 1.60 \times 10^{-19}}{4 \times 1.66 \times 10^{-27}} \times \frac{240 \times 1.66 \times 10^{-27}}{94 \times 1.60 \times 10^{-19}}$	C1
	ratio = 1.3	A1

37.

8(a)	antineutrino and positron both underlined (and no other particles)	B1
8(b)(i)	nucleon number = 27	A1
	proton number = 13	A1
8(b)(ii)	weak (nuclear force/interaction)	B1
8(b)(iii)	an (electron) antineutrino / $\bar{\nu}_{(e)}$ is produced (and this has energy)	B1
	X has kinetic energy	B1

38.

5(a)	$E = F / Q$	M1
	$F = ma$ and (so) $q/m = a/E$	A1
5(b)	$m = (4 \times 1.60 \times 10^{-19} \times 3.5 \times 10^4) / 1.5 \times 10^{12} (= 1.49 \times 10^{-26} \text{ kg})$	B1
	$= 1.49 \times 10^{-26} / 1.66 \times 10^{-27} = 9.0 \text{ (u)}$	A1
5(c)	protons: 4 and neutrons: 5	A1
5(d)(i)	nuclei have the same charge and so same (magnitudes of) force	B1
5(d)(ii)	nuclei have different masses and same force and so different (magnitudes of) acceleration	B1

39.

7(a)	nucleus is charged	B1
	the mass is <u>concentrated</u> in (very small) nucleus	B1
	or the <u>majority</u> of the mass is in (very small) nucleus	
7(b)(i)	$-(1/3)e$	B1
7(b)(ii)	$2q - (1/3)e = e$ so $q = (2/3)e$	M1
	up / u (quarks) (allow charm or top quarks)	A1

40.

6(a)	path/direction in which a (free) <u>positive</u> charge will move	B1
6(b)	(lines) closer together in Y/further apart in X	B1
6(c)(i)	$a = Eq / m$	C1
	or $F = Eq$ and $F = ma$	
	ratio = $(1e / 0.15 \text{ u}) \times (4 \text{ u} / 2e)$ or $1 / 0.15 \times 4 / 2$	C1
	ratio = 13	A1
6(c)(ii)	down quark charge is $-(1/3)e$	C1
	$-(1/3)e + q = -1e$ so $q = -(2/3)e$	A1
	$-(2/3)e$ is anti-up / $\bar{u}$ (quark) (allow charm or top antiquark)	B1

41.

7(a)	beta/ β	B1
7(b)	alpha/ α	B1
7(c)	gamma/ γ	B1
7(d)	beta/ β	B1

42.

7(a)(i)	proton number = 17 and nucleon number = 35	A1
7(a)(ii)	(electron) neutrino	B1
7(b)	d/down (quark charge) is $-\frac{1}{3}(e)$ or <u>two</u> d/down (quark charges) is $-\frac{2}{3}(e)$ or s/strange (quark charge) is $-\frac{1}{3}(e)$	C1
	charge = $-\frac{1}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e)$ $= -1(e)$	A1

43.

7(a)	number of protons = 92	A1
	number of neutrons = 142	A1
7(b)	$5.6 \text{ MeV} = 5.6 \times 1.60 \times 10^{-19} \times 10^6 (= 8.96 \times 10^{-13} \text{ J})$	C1
	number = $0.15 / (5.6 \times 1.60 \times 10^{-13})$ $= 1.7 \times 10^{11}$	A1
	or	
	$0.15 \text{ W} = 0.15 / (1.60 \times 10^{-19} \times 10^6) (= 9.38 \times 10^{11} \text{ MeV s}^{-1})$	(C1)
	number = $9.38 \times 10^{11} / 5.6$ $= 1.7 \times 10^{11}$	(A1)

44.

7(a)	number of protons = 95	A1
	number of neutrons = 146	A1
7(b)	Np/neptunium (nucleus) has <u>kinetic</u> energy or gamma/ γ -radiation produced	B1
7(c)(i)	$I = NQ / t$	C1
	$I = (6.9 \times 10^{11} \times 2 \times 1.60 \times 10^{-19}) / 30$ $= 7.4 \times 10^{-9} \text{ A}$	A1
7(c)(ii)	$P = (6.9 \times 10^{11} \times 5.5 \times 10^6 \times 1.60 \times 10^{-19}) / 30$	C1
	$= 0.020 \text{ W}$	A1

45.

7(a)(i)	most of the atom is empty space or the nucleus (volume) is very small <u>compared to the atom</u>	B1
7(a)(ii)	the nucleus is charged	B1
	the mass is <u>concentrated</u> in nucleus / small region / small volume / small core or the <u>majority</u> of the mass is in nucleus / small region / small volume / small core	B1
7(b)(i)	proton number = 84	A1
	nucleon number = 214	A1
7(b)(ii)	up down down changes to up up down / udd → uud or down changes to up / d → u	B1

46.

6(a)	the nucleus is charged	B1
	the majority of the mass (of atom) is in the nucleus	B1
6(b)	made up of quarks (so) not a fundamental particle	B1
6(c)	(Q =) $6.9 \times 10^{-9} \times 60$	C1
	number = $(6.9 \times 10^{-9} \times 60) / (2 \times 1.60 \times 10^{-19})$	C1
	= 1.3×10^{12}	A1

47.

6(a)(i)	up up down	B1
6(a)(ii)	up down down	B1
6(a)(iii)	(alpha-particle is) 2 protons and 2 neutrons	C1
	6 up, 6 down	A1
6(b)(i)	most of an <u>atom</u> is empty space or the nucleus (volume) is (very) small <u>compared with the atom</u>	B1
6(b)(ii)	the nucleus is charged	B1
	the majority of the mass of atom is in the nucleus	B1

48.

6(a)	α -particle mass given as 4u	B1											
	α -particle charge given as (+)2e	B1											
	both β -particles mass given as 0.0005 u	B1											
	β^+ charge given as (+)e and β^- charge given as -e (Completed table: <table border="1"> <tr> <td></td><td>mass / u</td><td>charge / e</td></tr> <tr> <td>α</td><td>4</td><td>(+)2</td></tr> <tr> <td>β^+</td><td>0.0005</td><td>(+)1</td></tr> <tr> <td>β^-</td><td>0.0005</td><td>-1</td></tr> </table>)		mass / u	charge / e	α	4	(+)2	β^+	0.0005	(+)1	β^-	0.0005	-1
	mass / u	charge / e											
α	4	(+)2											
β^+	0.0005	(+)1											
β^-	0.0005	-1											
6(b)(i)	neutron decays into proton and an electron / β^- particle	B1											

6(b)(ii)	down to up	B1
6(b)(iii)	(electron) antineutrino(s) emitted	B1
	energy (released in decay)/momentum shared between antineutrino and β^- particle	B1

49.

7(a)	(total) momentum before (decay) is zero or P has zero momentum	B1
	(total momentum after decay must be zero so) α -particle and Q have momenta in opposite directions (and therefore velocities are in opposite directions)	B1
7(b)	$p = 239 \text{ (u)} \times v$ or $4 \text{ (u)} \times 1.6 \times 10^7$	C1
	$239 \text{ (u)} \times v = 4 \text{ (u)} \times 1.6 \times 10^7$	A1
	$v = 2.7 \times 10^5 \text{ m s}^{-1}$	
7(c)	$E_{(K)} = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (1.6 \times 10^7)^2$	C1
	$= 8.5 \times 10^{-13} \text{ (J)}$	A1
	$= 8.5 \times 10^{-13} / 1.60 \times 10^{-13} \text{ (MeV)}$	
	$= 5.3 \text{ MeV}$	
7(d)(i)	1. R plotted at (95,147)	B1
	2. S plotted at (96,146)	B1
7(d)(ii)	(electron) <u>anti</u> neutrino	B1

50.

7(a)(i)	(electron) neutrino	B1
7(a)(ii)	nucleon number = 22	A1
	proton number = 10	A1
7(a)(iii)	up up down changes to up down down or up changes to down	B1
7(b)(i)	charge = $-\frac{2}{3}e$	A1
7(b)(ii)	antiup / anticharm / antitop	B1

51.

7(a)(i)	change in $A = 0$	A1
	change in $Z = (+)1$	A1
7(a)(ii)	(electron) <u>anti</u> neutrino	B1
7(b)(i)	up / u (charge) = $(+)\frac{2}{3}e$ or antidown / $\bar{d} = (+)\frac{1}{3}e$ or $(q) = \frac{2}{3}e + \frac{1}{3}e$	M1
	$q = (+)1e$	A1
7(b)(ii)	hadron(s)	B1
	meson(s)	B1

52.

7(a)(i)	P = 0 and Q = 137	A1
	R = -1 and S = 56	A1
7(a)(ii)	lepton(s)	B1
7(b)(i)	(charge of ddd / Y =) $-\frac{1}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e) = -1(e)$	B1
	(charge of $\bar{u}d$ / Z =) $-\frac{1}{3}(e) - \frac{2}{3}(e) = -1(e)$	B1
7(b)(ii)	meson: Z / $\bar{u}d$ because consists of a quark and an antiquark	B1
	baryon: Y / ddd because consists of three quarks	B1

53.

7(a)(i)	${}^{18}_9\text{F} \rightarrow {}^{18}_8\text{O} + {}^0_{(+1)}\beta^{(+)} + {}^{(0)}_{(0)}\nu(e)$ v or neutrino (B1) ${}^0_{(+1)}\beta^{(+)}$ (B1) ${}^{18}_8\text{O}$ (B1)	B3
7(a)(ii)	up quark to down quark	B1
7(b)(i)	must be three (anti)quarks as largest (negative) quark charge is $(-)\frac{2}{3}(e)$ or mesons can only have a charge of 0 or $\pm 1(e)$	M1
	(so hadron is) a baryon	A1
7(b)(ii)	any combination of three from: antiup (quark) / up antiquark and/or anticharm (quark) / charm antiquark and/or antitop (quark) / top antiquark	B1

54.

6(a)	particle with no internal structure / particle which cannot be broken down into anything smaller	A1
6(b)	charges: $u = (+)\frac{2}{3}(e)$ or $d = -\frac{1}{3}(e)$ or $s = -\frac{1}{3}(e)$	C1
	$(+)\frac{2}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e) = 0(e)$	A1
6(c)(i)	- same/equal mass - same/equal (magnitude of) charge - both fundamental (particles) - opposite (sign of) charge - one is matter and the other is antimatter Any two points, 1 mark each.	B2
6(c)(ii)	neutron/baryon consists of three quarks	B1
	pion/meson consists of one quark and one antiquark	B1

55.

7(a)	92 protons and 146 neutrons (in nucleus)	B1
	92 (orbital) electrons	B1
7(b)	charge = $2e$ (= $2 \times 1.60 \times 10^{-19}$ C)	C1
	mass = 4u (= $4 \times 1.66 \times 10^{-27}$ kg)	C1
	ratio = $(2 \times 1.60 \times 10^{-19}) / (4 \times 1.66 \times 10^{-27})$ = 4.8×10^7 C kg ⁻¹	A1
7(c)(i)	up down down / udd	B1
7(c)(ii)	up up up / uuu	B1

56.

6(a)	${}^{14}_7\text{X}$	B1
	${}^0_{-1}\text{e}^-$	B1
6(b)(i)	$d \rightarrow u + e^- + \bar{\nu}$ or $udd \rightarrow uud + e^- + \bar{\nu}$	B1
6(b)(ii)	$-1/3 (e) = +2/3 (e) - 1(e) (+0)$ or $2/3 (e) - 1/3 (e) - 1/3 (e) = 2/3 (e) + 2/3 (e) - 1/3 (e) - 1 (e) (+0)$	B1
6(c)(i)	electrons / β -particles (emitted from the nucleus) have a (continuous) range of / different (kinetic) energies	B1
6(c)(ii)	the (emitted) neutrinos take varying amounts of the (same total) energy (released in the decay)	B1

57.

7(a)(i)	X has same number of protons as Y (and so) charge of X is the same as the charge of Y	B1
7(a)(ii)	X has (one) more proton (than Z)	M1
	(so) X has greater charge (than Z)	A1
7(b)(i)	meson(s)	B1
7(b)(ii)	one quark and one antiquark	B1

58.

8(a)(i)	Y and Z have equal numbers of protons and (so) they have the same charge	B1
8(a)(ii)	Y has (two) fewer protons (than X)	M1
	(so) Y has less charge (than X)	A1
8(b)	(total) momentum before decay is zero or X has zero / no momentum	B1
	(total momentum after decay must be zero so) Y must have equal (and opposite) momentum to α -particle (so cannot be stationary / must have speed/velocity)	B1

59.

8(a)(i)	number of protons: equal/same	B1
	number of neutrons: unequal/different	B1
8(a)(ii)	down (quark) changes to up (quark) or up down down (quarks) change to up up down (quarks)	B1
8(a)(iii)	(electron) antineutrino	B1
8(b)	charm (quark charge) is $(+)2/3(e)$ or 2 charm (quark charges) is $(+)4/3(e)$ or bottom (quark charge) is $-1/3(e)$	C1
	charge = $+2/3(e) + 2/3(e) - 1/3(e)$ = $(+)1(e)$	A1

60.

7(a)	down charge = $-1/3(e)$ and charm charge = $(+)2/3(e)$	B1
	all antiquarks have opposite sign and same (non-zero) magnitude of charge as the corresponding quarks	B1
7(b)(i)	udd or cdd	B1
7(b)(ii)	$u\bar{d}$ or $c\bar{d}$	B1
7(c)(i)	lepton(s)	B1
7(c)(ii)	positron / neutrino / antineutrino	B1

61.

7(a)	a (very) small proportion of (alpha) particles are deflected (back) through large angles / angles greater than 90°	B1
	a large proportion of (alpha) particles pass straight through / deflected by small angles	B1
7(b)	electromagnetic wave / electromagnetic radiation	A1
7(c)	neutrino classified as a lepton only <u>and</u> positron classified as a lepton only	B1
	neutron classified as a baryon and a hadron and not as a lepton	B1
7(d)(i)	1.32 u	A1
7(d)(ii)	working states or implies 3 quarks and each quark has a charge of $(+) \frac{2}{3}(e)$	B1
	any combination of 3 quarks comprised of one or more of up / charm / top	B1

62.

7(a)	only baryon and nucleon underlined	B1
7(b)	number of α -particles = 6	B1
	number of β^- particles = 4	B1
7(c)	charm antiquark (charge) = $-2/3(e)$ or $-2/3(e) + q = -1(e)$	B1
	(other quark $q = -1/3(e)$ so) down / strange / bottom	B1

63.

8(a)(i)	23: nucleon number or number of neutrons and protons	B1
	12: proton number or number of protons	B1
8(a)(ii)	both numbers correct for X: $^{23}_{11}\text{X}$	B1
	both numbers correct for beta-plus: $^0_1\beta^+$	B1
8(a)(iii)	lepton(s)	A1
8(b)	(the emitted particles) have a single (kinetic) energy or beta particles / decay have a (continuous) range of (kinetic) energies	B1